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INTERNSHIP REPORT

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2 months research visit of Juliette Godet (3rd year PhD student at Université Gustave Eiffel, France) at NIWA Christchurch, New Zealand, under the supervision of Céline Cattoën-Gilbert. Funded by Université Gustave Eiffel.

GRAPHICAL ABSTRACT

Ensemble-based impact forecasting: Application and validation on the Westport 2021 flood

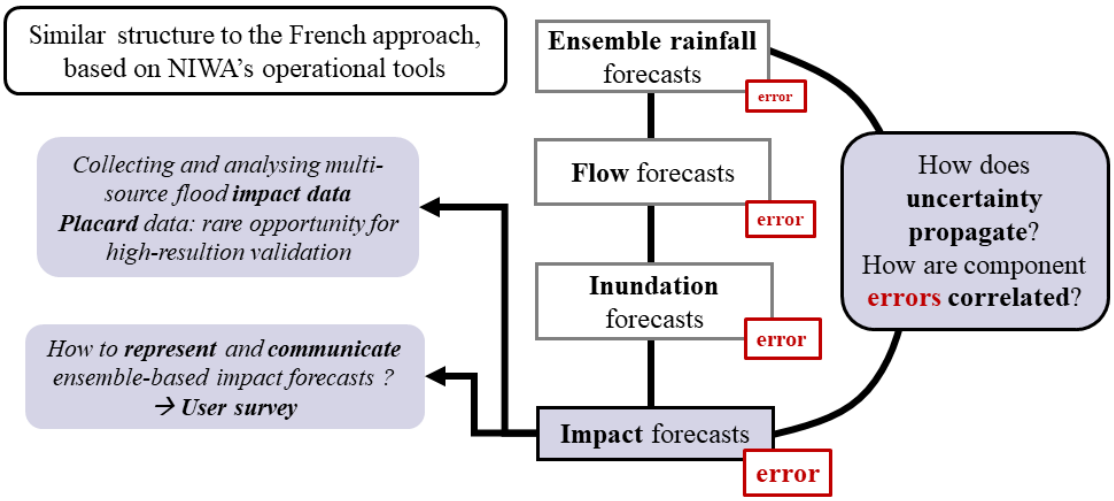


TABLE OF CONTENTS

1 Context **3**
1.1 NIWA and Université Gustave Eiffel . . . 3
1.2 PhD work and objectives of the research visit . . . 3
1.3 Some context regarding flood forecasting in Aotearoa VS France 4

2 Study case and forecast data **5**
2.1 July 2021 Westport flood event . . . 5
2.2 Rainfall Forecasts . . . 7
2.3 Flow Forecasts . . . 7
2.4 Inundation forecasts . . . 9

3 Collecting impact observations **11**
3.1 Number of claims from ICNZ . . . 12
3.2 Placard data . . . 12
3.3 Other leads . . . 14

4 Impact modelling **19**
4.1 RiskScape . . . 19
4.2 Binary impact detection . . . 20
4.3 Multi-category damage detection . . . 22

5 Cascading uncertainties **24**
5.1 Forecast performance through the chain . . . 24
5.2 Run-to-run jumps in forecast performance . . . 27
5.3 Correlation of Errors . . . 28

6 Comparing quantile-based approaches **32**

7 Flood library and interpolation methods **33**

8 Representing impact-based flood forecasts: user insights **35**
8.1 Questions and answers from the survey . . . 36
8.2 Analysis of the results of the survey . . . 45

9 Conclusions and perspectives **48**

References **49**

1 Context

Timeline of the research visit

Juliette Godet and Céline Cattoën-Gilbert met at EGU 2023. In August 2024, J.G reached out to explore the possibility of a short research visit in 2025. An application for funding was submitted to Université Gustave Eiffel in October 2024, accepted in January 2025. The research visit took place from March 10 to May 2, 2025.

1.1 NIWA and Université Gustave Eiffel

- NIWA represents New Zealand within the World Meteorological Organization (WMO) on hydrology-related matters. One of its missions is to develop tools that strengthen New Zealand's resilience to natural hazards, including floods. MetService (Meteorological Service of New Zealand), currently in charge of issuing weather warnings, will soon be merged with NIWA according to a very recent announcement, which will enhance collaboration between the two institutions.

- Université Gustave Eiffel is actively involved in flood risk management in France. It collaborates closely with SCHAPI (the national flood forecasting and warning service) to develop and improve flood forecasting tools, particularly for flash floods. The university also contributes to national research programs aimed at increasing the country's resilience to natural hazards.

This scientific visit brings mutual benefits. J.G broadens her expertise in international flood forecasting and enrich her PhD with a New Zealand case study. NIWA benefits from the evaluation methods developed during her PhD and gain new insights into its impact-based forecasts. The visit also helps establishing lasting collaboration between NIWA and Université Gustave Eiffel, who share strong research interests in flash flood forecasting, impact modelling, and uncertainty representation.

1.2 PhD work and objectives of the research visit

The French flash flood warning system, Vigicrues Flash, shares significant similarities with NIWA's national flood forecasting system used in New Zealand [Cattoën et al., 2022]. Both are designed to provide early information on floods in small, ungauged catchments, and they also share a major limitation: the warnings issued are based solely on the predicted flood intensity, without taking into account the potential damages these floods could cause—damages that strongly depend on the exposed assets and local context.

Juliette Godet's PhD research aims to explore how accounting for territorial exposure and associated impacts (the concept of impact-based forecasting) can be integrated into such flood warning systems, to improve the relevance and usefulness of the alerts issued. The initial case study focused on the French Mediterranean region, using a modelling chain similar to the one employed in Vigicrues Flash. This chain was extended to include flood extent mapping and identification of potentially affected assets, and was evaluated based on its ability to represent the real-world impacts observed over a long historical period (about 10 years). New evaluation approaches had to be developed, drawing on diverse sources of impact data, such as insurance claim records, official natural disaster declarations, and emergency response reports.

The goal of this scientific visit is to investigate the implementation and assess the relevance of impact-based warnings in the context of the flood warning system developed by NIWA in New

Zealand, building on the methodologies developed during Juliette Godet's PhD. This involves developing new modelling chains based on NIWA's operational tools, and adapting the evaluation strategies to reflect the specific characteristics of the New Zealand context and the available impact datasets, in order to model and estimate real-world flood impacts.

1.3 Some context regarding flood forecasting in Aotearoa VS France

- **Aotearoa**: There is no national agency tasked with coordinating hydrological flood warnings or overseeing hydrometric monitoring. Instead, flood warnings are issued by regional or local authorities, reflecting a decentralized system where decisions are made by those closest to the affected areas and most familiar with local conditions. At the national level, MetService (the Meteorological Service of New Zealand Ltd.) is responsible for issuing severe (orange) and extreme (red) weather warnings across land, sea, and air. Once these weather inputs are provided, the responsibility for flood forecasting and warning falls to regional councils. These councils rely on their own models—often developed with support from engineering consultancies—and there is no standardized flood modeling approach used across the country. Importantly, there is generally no further operational coordination with MetService at this stage; MetService does not retain visibility over the flood forecasts and typically has no access to river flow data.

- **France**: In France, hydrological flood warnings and hydrometric monitoring are centrally coordinated by national agencies. The responsibility for issuing flood warnings lies with the Vigicrues network, operated by regional flood forecasting centers under the oversight of the national agency SCHAPI (Service Central d'Hydrométéorologie et d'Appui à la Prévision des Inondations). These centers monitor river conditions and issue warnings based on a standardized color-coded system. In parallel, Météo-France, the national meteorological service, issues weather warnings at the département level for hazards such as heavy rain, wind, or snow. While the warnings issued by both services are distinct and shown on separate maps, they collaborate closely (the warning level colours are identical, the flood and weather operational forecasting rooms are literally side by side, and the websites revealing the warning maps are connected).

2 Study case and forecast data

At the beginning of the internship, several flood events were considered as potential case studies, including:

- A series of flood events that affected the town of Westport in 2021 and 2022. This included the July 2021 flood (which caused severe damage), the February 2022 flood (which caused light damage), and the August 2022 flood (which caused no significant damage). NIWA had previously undertaken a project in Westport aimed at implementing a real-time flood model capable of producing inundation maps. As a result, reforecasts of inundation are available for all three events.
- Cyclone Gabrielle in February 2023, a major extreme weather event that struck the North Island of Aotearoa New Zealand, with an estimated total cost of NZD 14.5 billion.

However, due to time constraints and the limited availability of relevant data, the internship ultimately focused on the July 2021 Westport flood event. The following sections present an overview of the event and describe the data provided by NIWA for this study.

2.1 July 2021 Westport flood event

Brief description of the event, *extracted from Johnson et al. [2025]*

> *"The July 2021 flood in Westport lasted 72 hours from 15th to 18th July. The event was characterized by heavy rainfall and a rapid rise in river levels that caused the Buller River to breach its banks. This caused significant flooding across the Buller District and necessitated the evacuation of 826 properties and over 2,064 people in Westport. Bridges were closed, cutting access to Westport. The flood caused substantial damage, with 71 homes deemed unsafe for habitation and 388 homes requiring assessment for damage. A total of NZ$97.2 million in insurance claims was paid to Westport residents."*

Meteorological drivers, *extracted from Cattoën et al. [2022]*

> *"A pulse of tropical air originating over the Indian Ocean combined with an upper-level trough and front delivered a humid and gusty northerly flow to the central and northern South Island. Within 72 h, rainfall stations in the Southern Alps recorded 697 mm at Ivory Glacier, 413 mm at Arthurs Pass"*

Hydrological reaction, *extracted from NIWA [2021] and Johnson et al. [2025] and Cattoën et al. [2022]*

> - *"The water level at Te Kuha peaked at 12.8 metres at 1.35pm on Saturday, July 17, several hours ahead of the peak flooding in Westport. In the 1970 flood the flood peak was 11.9 metres. This was the biggest flood recorded since the Te Kuha monitoring site was established in 1963."*
> - *"We can confidently say that the recent flood flows in the Buller River are the highest of any river in New Zealand since 1926."*
> - *"The predicted peak flow of the probabilistic forecasts (median of 8,564 m3/s on 17th July) was close to the actual peak flow (8,890 m3/s)."*
> - *"As bridges closed, the NIWA field team measured 7640 m3/s (for a catchment of 6350 km2 and 454 m^3/s mean flow)—the largest direct streamflow measurement in NZ"*

Figure 1: Westport, July 2021. ©NZ Defence Force

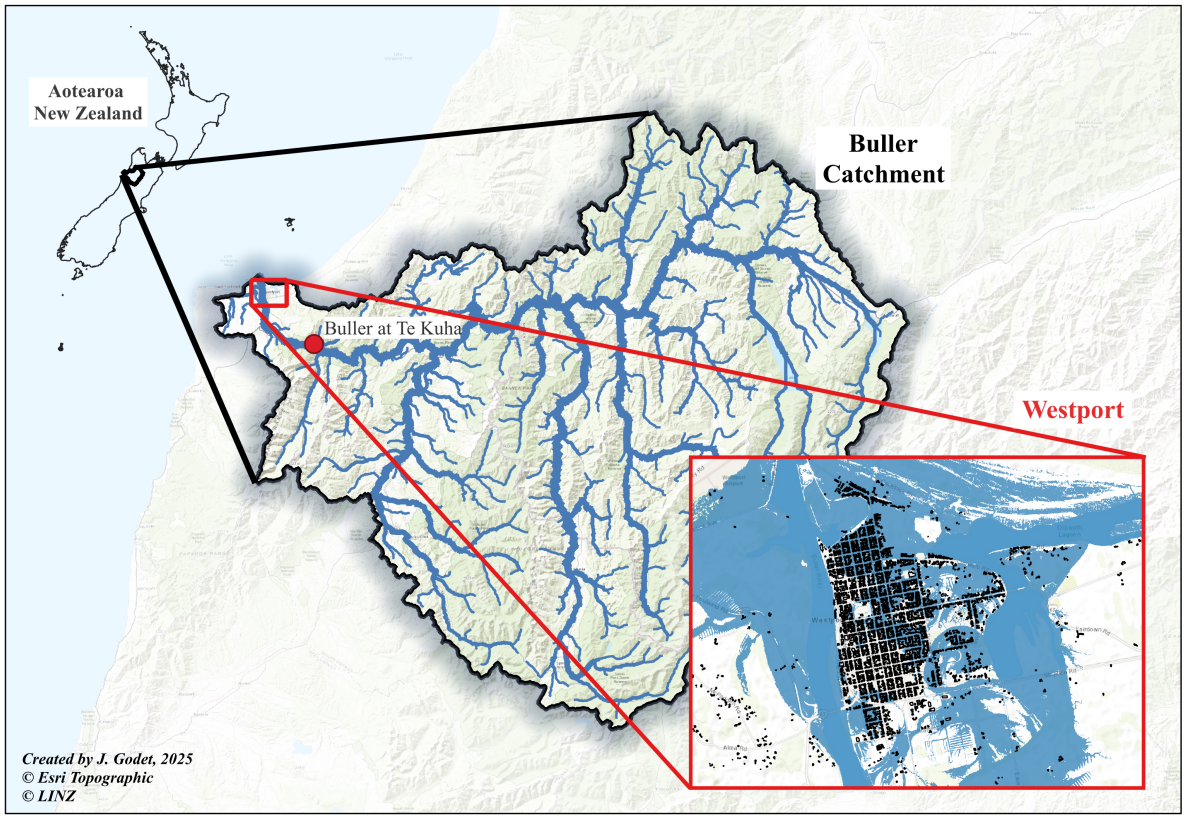


Figure 2: Presentation of the Buller catchment and Westport, with July 2021 reconstructed flood inundation map (pixels with water depths > 5cm are coloured in blue)

Comparison with Mediterranean Flash Floods

The July 2021 flood event in Westport is markedly different from the Mediterranean flash floods studied in the context of this PhD in France.

- The duration differs: this was not a flash flood with a rapid rise and recession within 24 hours, but rather a prolonged event that lasted several days.

- The flood mechanisms also differ: although this event was primarily riverine, coastal processes play a significant role in Westport's flood dynamics.
- However, some meteorological drivers are comparable: the Buller catchment features an even greater elevation range than typical Mediterranean catchments, and both regions are primarily fed by orographic rainfall.
- The exceptional impacts recorded during this event resonate with the high-impact floods frequently observed in the French Mediterranean region.

2.2 Rainfall Forecasts

In this study we used the NZens ensemble rainfall product, which consists of 18 members with a run length of 5 days. Forecasts are issued twice daily (at the time of the 2021 event; however more recent versions support up to four runs per day). The gridded rainfall resolution is 4.4 km. NZens is based on the RAL1-M regional numerical weather prediction model developed in the UK [Bush et al., 2020].

NZens is a dynamical, convective-scale ensemble model and differs from NZCSM (New Zealand Convective Scale Model), which is used in NIWA's operational flood forecasting system, AFAS [Cattoën et al., 2022]. Unlike NZens, NZCSM is a deterministic model that has been statistically perturbed in terms of rainfall, soil moisture, and baseflow to account for uncertainty in streamflow forecasts.

Rainfall Radar: France vs. New Zealand

Rainfall forecasts used for flood prediction in New Zealand rely on global or regional numerical models. Indeed, radar coverage differs significantly between France and New Zealand. In New Zealand—particularly on the West Coast of the South Island—gaps in radar coverage make it challenging to obtain reliable rainfall observations, especially at high spatial and temporal resolutions.

Lead Times – Nowcasting vs. Forecasting

As previously mentioned, NZens forecasts have a run length of 5 days. This contrasts sharply with the ensemble rainfall forecasts used in Vigicrues Flash in earlier work [Godet et al., 2023], where the maximum lead time was just 6 hours. In this context, the term "forecasting" is more appropriate, while "nowcasting" better describes the approach used in France.

For the validation of the rainfall forecasts, catchment-averaged rainfall was computed for each time step using the "best" gridded rainfall product, which is based on observations. This variable, referred to as *aprecip*, will be used in the validation process as described in later sections.

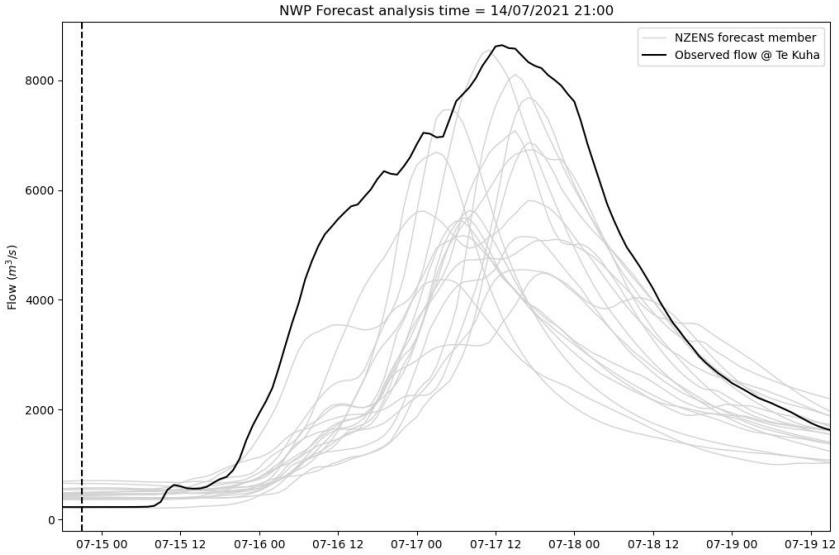
2.3 Flow Forecasts

Flow forecasts were produced using the TopNet hydrological model [Clark et al., 2008]. TopNet is a continuous, semi-distributed, physically-based hydrological model that operates on an hourly time step. It combines a water balance model applied to each subcatchment with a streamflow routing component using the kinematic wave model. TopNet is used operationally within NIWA's national flood forecasting system, AFAS, as described in [Cattoën et al., 2022]. It has been developed internally by NIWA and is not currently open source.

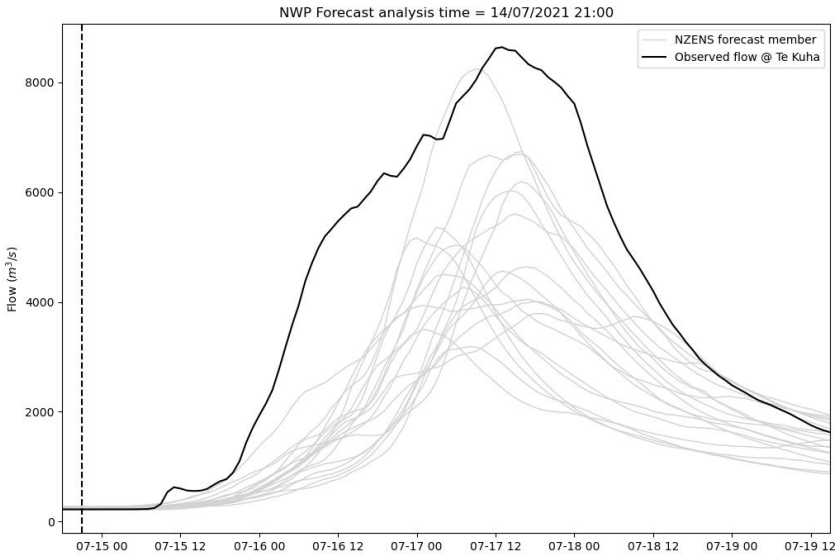
TopNet vs. CINECAR

At first glance, TopNet may appear similar to the CINECAR model developed by the Université Gustave Eiffel team [Gaume et al., 2004, Naulin et al., 2013]. However, CINECAR is originally an event-based model, although recent research has begun to explore coupling its vectorial routing module with the continuous SMASH production module [Colleoni and Huynh, 2024].

Initially, flow forecasts were generated using an older version of TopNet, as it was immediately available. Updated forecasts using the newer Strahler 2 version of TopNet were obtained two weeks after the start of the internship. Two versions were provided: with and without river data assimilation (see Figure 3). Both versions significantly underestimated river discharge, though the underestimation was less pronounced in the version without data assimilation. Therefore, this version is the focus of the remainder of the analysis.



(a) Without river data assimilation



(b) With river data assimilation

Figure 3: Different versions of provided forecasts - with and without river data assimilation

We focus on the flow forecasts at the Buller River at the Te Kuha station, which is the closest hydrometric station to the town of Westport. Using observed discharge data, the TopNet model was calibrated on the July 2021 event, allowing for the reconstruction of an observed

hydrograph. This hydrograph, shown in Figure 3, serves as a reference and will be used for validation purposes in the subsequent sections of this report.

Similar validation data for France

This compares with the hydrographs reconstructed with Cinecar for the Aude2018, Alpes-Maritimes2015 and Argens2010 events that are provided in Godet et al. [2025].

2.4 Inundation forecasts

2.4.1 Flood library

As mentioned earlier, NIWA has developed a flood model for Westport. It is based on a flood library of approximately one hundred scenarios, each representing a unique combination of two parameters considered to be the main drivers of flooding in the area:

- Peak discharge at the Te Kuha station.
- Peak sea level.

All the scenarios in this 2D parameter space are shown in Figure 4.

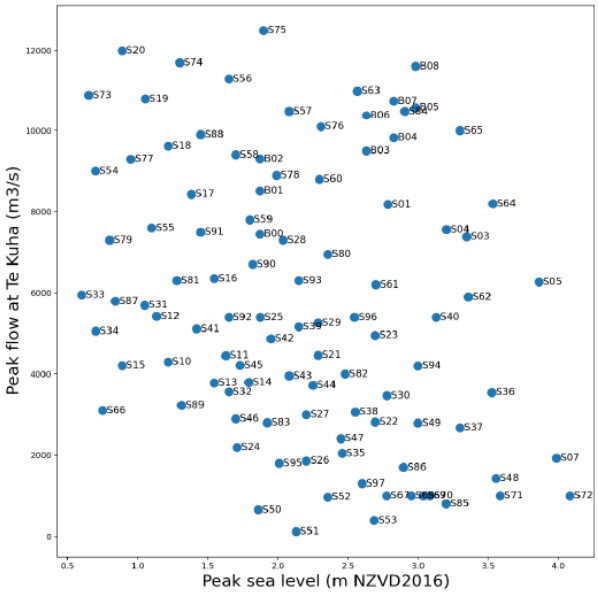


Figure 4: Precomputed flood library ©C. Cattoën-Gilbert

For research purposes, the NIWA team investigated how to construct a reduced flood library under resource constraints—e.g., if only 15 scenarios could be implemented. They compared different selection methods, including an "Holistic" approach, a "Maximum Dissimilarity / Uniform" approach, and a "Random" approach. A scientific paper detailing this work will be submitted soon by the NIWA team.

Toward a Mediterranean-Equivalent Flood Library

One of the approaches explored in the Westport study is a "Design Library" method, consisting of 15 scenarios under different climate change conditions (present, RCP 4.5, 6.0, and 8.5), various flow return periods at Te Kuha (20, 50, and 100 years), and different peak sea levels. However, since the July 2021 flood was primarily driven by river discharge, and to enable comparison with the French case study, it is possible to consider only the 1D peak flow dimension. This results in a simplified library of 9 scenarios, summarized in Table 1.

	RP = 20y	RP = 50y	RP = 100y
present conditions	7451	8514	9304
RCP 4.5		9493	10383
RCP 6		9818	10738
RCP 8.5		10564	11582

Table 1: Flow values (m^3/s) of Te Kuha in the 1D Design flood library

In the French flood library, eight monofrequency scenarios were used, corresponding to the 2-, 5-, 10-, 20-, 50-, 100-, 500-, and 1000-year return periods of peak discharge.

However, to remain consistent with the model delivered to the Westport stakeholders, the results presented in this report are based on the full 100-scenario flood library, with flood maps interpolated using the TIN method. This point will be revisited in Section 7.

Hydraulics Computations

For both the Mediterranean and Westport flood libraries, hydraulic simulations were carried out using a 2D model based on the Saint Venant shallow water equations. However, while the Westport model was run in unsteady mode, using full hydrographs to capture the temporal dynamics of the flood, the Mediterranean simulations were conducted under steady-state assumptions, using only the peak discharge associated with each return period scenario.

2.4.2 Interpolation method

In order to reconstruct flood extents in real time based on the flood library, NIWA has explored several interpolation methods:

- Nearest-neighbour: selecting the closest scenario in the 2D parameter space without interpolation.
- TIN (Triangular Irregular Network): a 2D linear interpolation method that divides the standardized parameter space into triangles. The forecast lies within one of these triangles, and interpolation is performed based on the values at the three vertices.
- AI-based interpolation: using a Random Forest algorithm.

According to comparison results, which will be discussed in NIWA's forthcoming publication, the spread of differences between methods narrows as the size of the flood library increases. Furthermore, the AI-based method does not significantly outperform the simpler interpolation approaches.

Comparison with the flood map reconstruction method used in France

In the PhD work focused on Mediterranean catchments, flood maps are reconstructed in real time by selecting the closest scenario for each subcatchment—i.e., no interpolation.

Given that the average size of a Mediterranean subcatchment in this study is 4.2 km^2, which is comparable to the size of Westport's city center, the corresponding method would be equivalent to a 1D nearest-neighbour approach, relying solely on the peak discharge at Te Kuha.

2.4.3 Reference flood map

For the July 2021 Westport event, the NIWA team has reconstructed a water level flood map based on collected high water marks, local knowledge and hydraulics simulations. It is shown in figure 5.

Similar validation data for France

This compares with the reconstructed flood extents and collected High Water Marks for the Aude2018, AlpesMaritimes2015 and Argens2010 events that are provided in Godet et al. [2025].

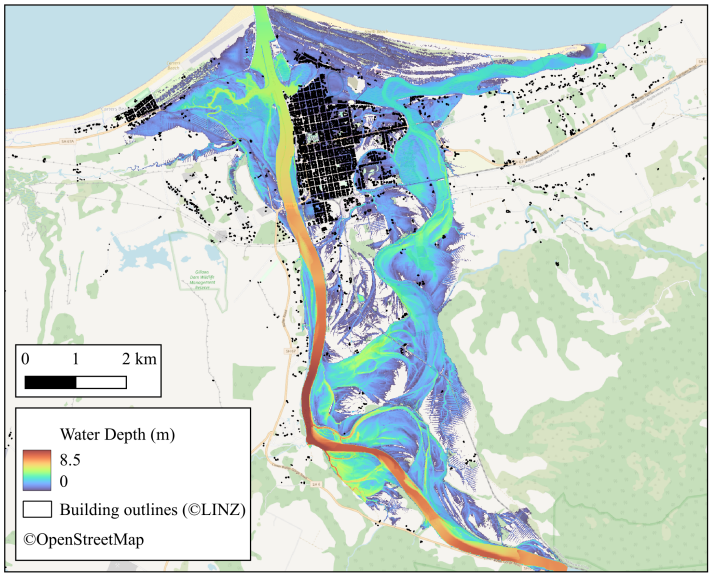


Figure 5: Reconstructed flood map

3 Collecting impact observations

The objective of this work is to conduct a block-by-block validation of the flood forecasting chain. The previous sections presented the simulation outputs and validation data available for the existing components of the chain: rainfall, streamflow, and inundation. The next step is to integrate an impact component, which poses the significant challenge of obtaining reliable impact observations [Harrison et al., 2021, 2022a,b, 2024]. This section outlines the various datasets that were considered for this purpose.

From broad-scale to high-resolution

In France, the objective was to collect flood impact data over a broad spatio-temporal scale—covering the Mediterranean region over a continuous 10-year period. While a similar approach would be ideal for New Zealand, particularly to support future research, the present study realistically focuses on the July 2021 flood in Westport. This means that the analysis is limited to a single event and a single city. However, this narrower scope allows for a more detailed investigation than what was done in France, particularly in terms of spatial resolution.

3.1 Number of claims from ICNZ

The Insurance Council of New Zealand (ICNZ) has been documenting the economic costs of natural disaster events since 1968. These records detail the costs borne by the insurance industry in settling claims for damage resulting from natural hazards such as floods, storms, wind, and earthquakes. They can be found at https://www.icnz.org.nz/industry/cost-of-natural-disasters/.

The information provided by ICNZ for the "West Coast Flooding" event, which occurred between July 16 and July 19, 2021, is presented in Table 2. Given that the majority of the impacts were concentrated in the town of Westport, it is reasonable to assume that the reconstructed flood map shown in Figure 5 encompasses the full extent of the insured losses reported in Table 2.

Inflation-adjusted cost (2023): 110.2 million NZD

Claim Type	**Number of Claims**	**Insured Losses (NZD)**
House & Contents	1,368	87,921,014
Commercial	182	5,882,377
Marine	2	51,823
Motor	307	3,130,708
Crops	0	0
Other	5	239,763
Total	**1,864**	**97,225,685**

Table 2: Number of claims and insured losses by claim type for the Westport July 2021 flood.

As of today, the ICNZ database includes 229 events. However, no information was found on the criteria used to include an event in the database (e.g., cost threshold).

French datasets with comparable content and resolution

Similar data sources available in France include:

- For detailed claim counts: the iCatNat database, which provides estimates of affected buildings at the municipality level.
- For insured loss estimates: the event library of the Caisse Centrale de Réassurance (CCR), the public reinsurer offering state-backed coverage for natural disasters. Their portal (https://catastrophes-naturelles.ccr.fr/les-evenements) provides cost ranges for events with insured losses exceeding €10 million.

Similar data, although not publicly available, was provided by CCR specifically for the purposes of this PhD. It consists of a ranking of affected river reaches (with an average size of 4km², thus offering a higher resolution than municipality- or event-scale data) based on the number of insurance claims per flood event. However, CCR did not provide the exact number of claims nor the insured loss amounts.

3.2 Placard data

In New Zealand, residential properties affected by a natural disaster are assessed using a placard system managed under the Building Act. This system classifies building usability based on observed damage, with three categories: white placards (safe to occupy), yellow placards (restricted use), and red placards (unsafe to enter).

Two datasets were provided by the West Coast Regional Council:

- One corresponding to the initial inspections carried out by Urban Search and Rescue (USAR), which involved surveying all dwellings—including both affected and unaffected ones—and assigning a damage category (see Table 3).
- The other corresponding to the final placard assessments conducted by MBIE (Ministry of Business, Innovation and Employment). These assessments focused on the subset of buildings identified by USAR as affected. In addition to placard assignment, MBIE assessors also recorded a damage category (see Table 3).

While both datasets initially contained a large number of attributes (which differed between the two), for the purpose of this study, we retained the damage category, as it was the most relevant variable and the only one common to both datasets. This enabled us to merge them into a single dataset, summarized in Table 3 and spatially represented in Figure 6.

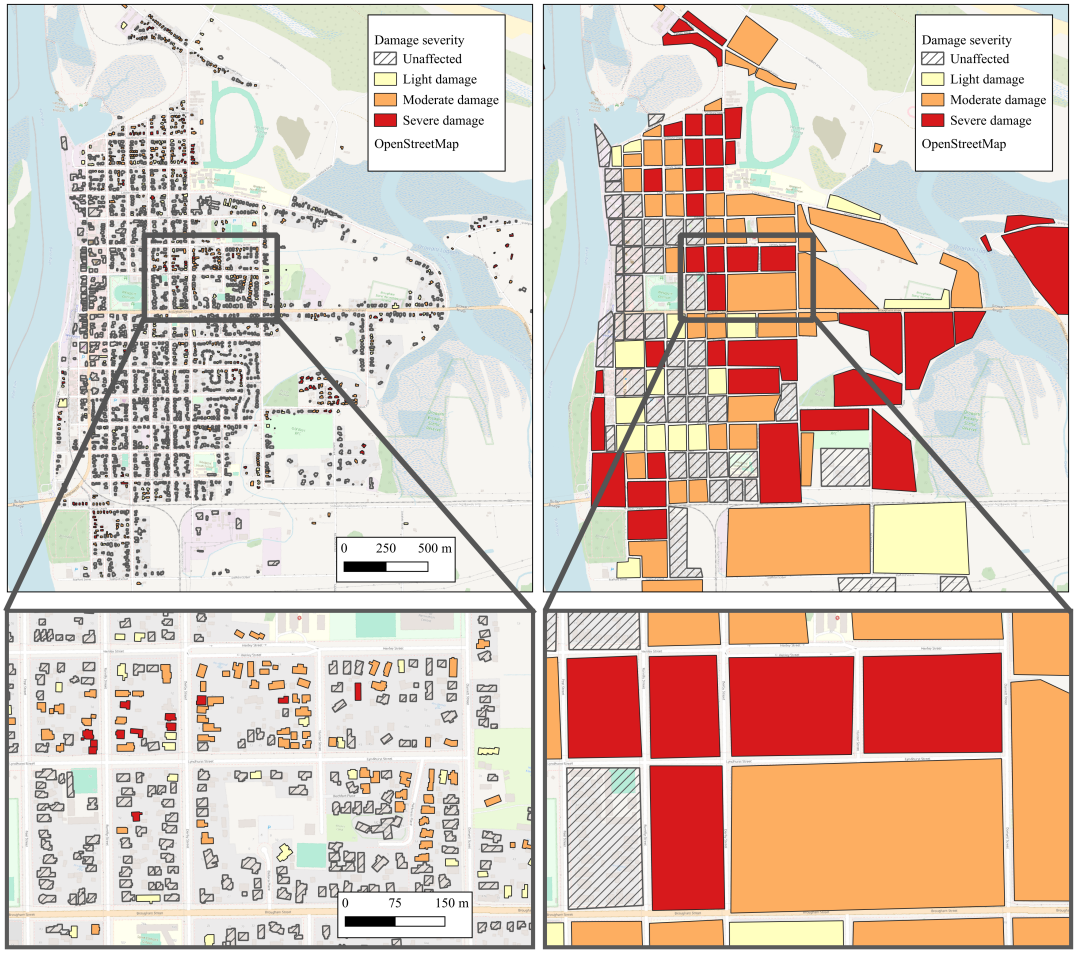


Figure 6: Damage severity at the building and block scales according to the placard data for 2021 July Westport event.

Figure 6 shows that not all buildings in Westport are included in the placard dataset. In some cases, this is because the dataset focuses specifically on residential buildings, and some excluded structures are non-residential. However, manual checks using Google Maps suggest that some excluded buildings do appear to be residential. This indicates that the placard dataset is not fully exhaustive. As a precaution, the analysis comparing simulated impacts to this validation dataset will be restricted to the subset of Westport buildings that are covered by the placard data.

Figure 6 also illustrates what the dataset looks like when aggregated at a "block" scale (manually defined areas regrouping, on average, 15 buildings). In this example, aggregation was performed by retaining the maximum damage severity observed within each block. Alter-

damage	source	building_id	DamageNb
NodamageUnaffected	USAR	2513599	0
Slightdamage	USAR	2513425	1
Moderatedamage	USAR	2513602	2
Severedamage	USAR	2513595	3
TotaldamageDestroyed	USAR	2513358	3
light	MBIE	2513596	1
moderate	MBIE	2513597	2
heavy	MBIE	2513604	3

Table 3: Illustration of the placard dataset and of all possible cases in the merging process, resulting in the "DamageNb" column.

natively, aggregation could be based on summing the DamageNb column from Table 3 within each block. This would produce a Damage Score that captures both the severity and the number of affected buildings, providing a richer (but more complex) representation of the impact.

Such aggregation is useful, as expecting a flood forecasting chain to deliver accurate damage predictions at the building level is extremely demanding—and arguably unrealistic. This limitation will be further discussed later in the report.

French datasets with comparable content and resolution

In France, no dataset exists that documents residential building damage with a similar level of geolocation accuracy, exhaustiveness, and damage detail. The closest available proxy is the database of fire service operations, which is highly detailed (time-stamped, precisely geolocated). However, it has several limitations:

- Not all recorded operations are related to building damage.
- The database does not include damage assessments.
- A single operation recorded at one location may actually involve multiple displacements and multiple buildings.

3.3 Other leads

Validation results based on the two aforementioned impact datasets only will be presented in this report. This section presents other datasets that were collected (but not used for various reasons) and datasets that were considered but could not have been collected.

3.3.1 Building impact survey

On August 10–12, 2021, 24 days after the July flood event, NIWA scientists conducted on-site residential building damage assessments on 247 buildings (see Figure 7). Their objective was to collect impact data to develop an AI-based model for damage prediction [Paulik et al., 2024]. To achieve this, they gathered information on flood hazard characteristics (e.g., water depth, flow velocity, debris) and building attributes (e.g., dwelling type, structural frame, floor height, foundation type, construction period, number of storeys, wall cladding).

This dataset was collected early during the internship timeline and was initially used to assess whether the forecasting chain could anticipate the buildings that were damaged as recorded in the dataset. However, its main limitation was its lack of exhaustiveness, which made it impossible to conduct a rigorous evaluation that would not only verify correct detection of impacts but also account for false alarms (i.e., predicted impacts where none actually occurred). As a result, this dataset was abandoned in favour of the placard dataset.

French datasets with comparable content and resolution

Several datasets sharing the same benefits (high level of detail, precise geolocation) and drawbacks (lack of exhaustiveness) have been collected for various flood events in the French Mediterranean region:

- **DamaGIS**: Manually compiled geodatabase of flood damages based on social media, local authorities, and field surveys.
- **Murex**: Online platform documenting flood damages through post-flood damage assessments.
- **Visov**: Database of flood impact reports collected through social media monitoring by volunteers.
- **Geolocated pictures**: Photo library capturing impacts such as damaged buildings, flooded vehicles, and devastated landscapes.

However, one difference is that these datasets do not focus exclusively on building damage, but also cover other types of impacts.

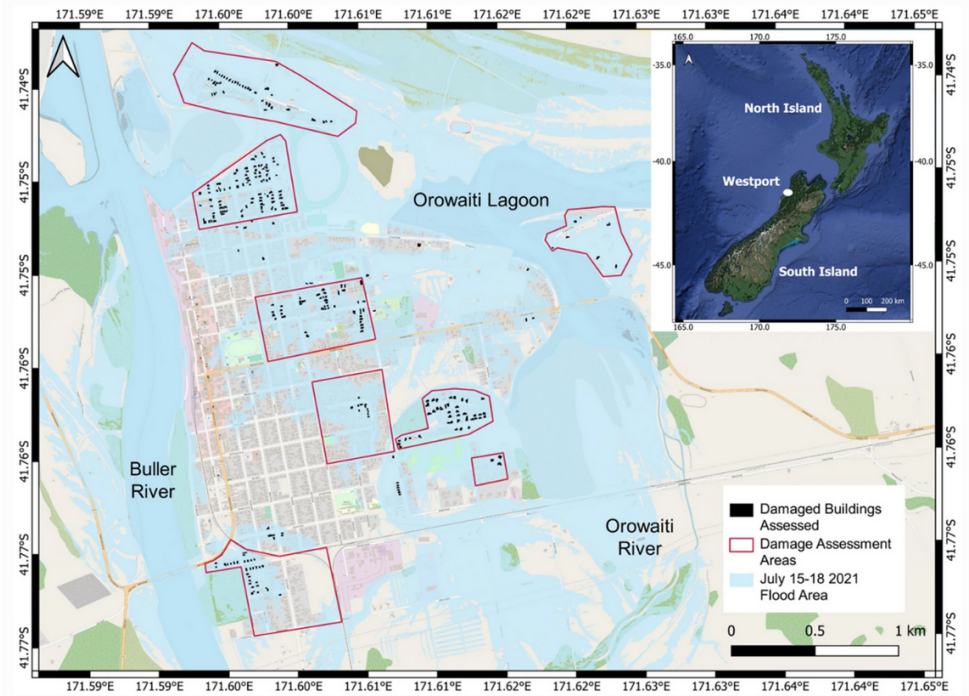


Figure 7: Residential building damage assessment locations in Westport for the 15–18 July 2021 flood event [Paulik et al., 2024].

3.3.2 Natural Hazard Commission

The Natural Hazards Portal, and particularly its claims map available at https://www.naturalhazardsportal.govt.nz/s/claims-map, is a highly valuable data source for natural hazard impacts. It provides data on insurance claims for residential property damage (buildings and land) caused by natural hazards in Aotearoa New Zealand. These claims are lodged with the Natural Hazards Commission (NHC), which provides insurance cover for homeowners whose properties are damaged by a natural hazard, provided their home insurance policy includes fire insurance (which most policies do).

However, unlike hazards such as landslides, earthquakes, tsunamis, and volcanic activity, the Natural Hazards Commission's cover for storms and floods is limited to certain areas of land under and around the insured home — land cover only. Any damage to the home itself must be covered through the homeowner's private insurance policy.

Specifically, in the case of floods:

- NHC provides some cover for damage to land:
  - under the home,
  - under insured separate buildings and structures,
  - within 8 metres of the home or insured separate buildings and structures,
  - under or supporting the main accessway to the home (e.g., a driveway), up to 60 metres from the home.
- NHC provides some cover for bridges and culverts:
  - located on the land within 8 metres of the home or insured structures, or
  - located under or supporting the main accessway, up to 60 metres from the home.
- NHC provides some cover for retaining walls:
  - that support or protect the home, insured structures, or insured land, and
  - are located within 60 metres of the home.

Because the Natural Hazards Commission (NHC) only provides cover for land damage in the case of floods, this database — although rich and exhaustive — is particularly tricky to use. On the one hand, it could be appropriate for validating forecasted building impacts, since in many cases, flooded properties also experience eligible land damage (e.g., debris deposits), which results in a claim. However, this is not systematic: a building can be affected without the associated land qualifying for cover, and homeowners may also be unaware of their eligibility and therefore not file a claim.

Use of insurance claim data for impact model validation

In France, we have been able to collect insurance claim data related to building damage. We have then based our validation on the assumption that for buildings where the water depth exceeded a certain threshold, there would be a claim, although this is not necessarily true, and for various reasons (e.g. deductible effect, precautionary measures, floor heights...).

The NHC database documents claims related to land damage only. Land damage is less directly correlated with water depth and would require much stronger assumptions for modelling. Nevertheless, it remains interesting to compare with impact forecasts, and the richness and exhaustiveness of this data source — available across all of New Zealand, over a long and continuous time scale — make it particularly valuable.

Late data

A bulk export of all flood-related claim data was requested from the NHC on March 13th. The contact point for this request was Geoffrey Spurr, Principal Loss Modelling Analyst at NHC. The data was received on May 7th, after the conclusion of the internship. It was shared for a specific use within the context of the current project and PhD research; therefore, NIWA will not be permitted to reuse it beyond the scope of this work. However, this initial collaboration has helped establish a clearer understanding of the data's potential value for flood modelling and may facilitate faster access in future requests.

A preliminary analysis of the dataset revealed that for the Westport 2021 floods, only five land claims were recorded in the NHC database—compared to 649 damaged buildings identified in the placard dataset.

3.3.3 NZ Historic Weather Events Catalogue

In the search for flood impact data on a large spatio-temporal scale (as conducted during the PhD work in France), the New Zealand Historic Weather Events Catalogue, developed by NIWA and available at https://hwe.niwa.co.nz/, was considered. This catalogue compiles major

weather events that caused significant damage or casualties over the past 200 years, from the 1800s to the present. Information has been gathered from a wide range of sources, including newspaper articles, journals, books, and databases contributed by various organisations and individuals. For each event, the catalogue records the affected regions, the types of hazards involved, and the resulting impacts.

Below is a summary of the information found in the catalogue for the July 2021 flood event:

📎 **Impacts of the July 2021 flood in the Buller district** *from* https://hwe.niwa.co.nz/event/July_2021_South_Island_Flooding

- The Buller District Council declared a State of Emergency, expecting that houses would be flooded.
- The Buller River was running at a 50-year flood level.
- 1000 people spent the night in evacuation centres in the Buller district.
- A Temporary Accommodation Service was set up to help those whose homes were damaged by the flooding and severe weather across the West Coast and Marlborough.
- The government pledged $300,000 to help the Buller district.
- About 1000 stock may have drowned in the flood event.
- Water conservation orders were issued for residents in Buller. They were asked to use water sparingly.
- The West Coast Civil Defence asked residents in particular streets to voluntarily evacuate. An evacuation centre was set up at Westport South School.
- The NZ Army helped to evacuate 200 Westport residents.
- Eighty-nine Westport homes were red-stickered, meaning they had suffered severe or irreparable damage, and 400 were yellow-stickered, meaning they had suffered moderate damage.

Although the Historic Weather Events Catalogue provides valuable information, it was not used in this study. First, it does not offer precisely geolocated data, which is essential for validating forecasted flood impacts at a fine spatial scale. Second, although aggregated data can still be useful — see section 3.1 — the catalogue does not provide appropriate quantitative data for the purpose of our study. The catalogue may still be useful for broader statistical analyses over long time periods, given its extensive temporal coverage. However, as it only records events associated with significant damage or casualties, threshold bias [Wyatt et al., 2023] will be a limit for systematic impact validation.

3.3.4 Extreme Weather Event database over Aotearoa New Zealand

The Aotearoa New Zealand (ANZ) Extreme Weather Events (EWE) database [Kreft et al., 2023] provides a comprehensive record of extreme weather events across ANZ, covering the period from March 1996 to December 2021. For each event, a score is assigned to each affected region, and this score is based on two components:

- The hazard type, labelled as **R** for broad-scale heavy rain events (over areas $> 1000km$), typically driven by synoptic features like frontal bands ; **C** for convective events, primarily thunderstorms, either embedded in frontal bands or isolated ; **W** for broad-scale strong wind events ; **S** for broad-scale heavy snow events ; **T** for extreme temperature events, which may be highly localized.
- An ”Extreme likelihood” rating (1 = high, 2 = moderate, 3 = low), assigned based on expert judgment and subjective evaluation.

The information in the EWE database was primarily sourced from MetService and from the aforementioned NZ Historic Weather Events Catalogue, supplemented by media reports and, where available, insured loss data from the Insurance Council of New Zealand (ICNZ).

The main limitation of the EWE database in the context of our work is that it focuses primarily on meteorological rather than hydrological events. Furthermore, as illustrated in table 4, the information recorded in the database is generic and impacts are not recorded in a structured way or always captured.

Attribute	Value
Unique Identifier	2021.52
Start Date	2021-07-15
Event Title	July 2021 Buller RED warning and extreme flooding
Source	MetService
Northland	
Auckland	
Waikato	3R
Bay of Plenty	3R
Gisborne	3R
Hawke's Bay	
Taranaki	3R;3W
Manawatu-Whanganui	3R
Wellington	3R;2W
Tasman/Nelson	1R
Marlborough	1R;3W
West Coast	1R
Canterbury	2R
Otago	
Southland	
Chatham Islands	
Comments	Widespread severe weather (Thu 15 to Sun 18); with Buller hit hardest by heavy rain and extreme flooding; but also impacting Marlborough; with thousands evacuated in Westport and Wairau River/Valley. Major road closures. Very heavy rain also in Westland; Canterbury; Tasman; Tararua Range; Taranaki; Bay of Plenty. Severe gales also impacted central NZ; with wind damage in Wellington/Wairarapa; power outages in Taranaki.
URL	https://hwe.niwa.co.nz/event/July_2021_South_Island_Flooding

Table 4: Details of the July 2021 Buller RED warning and extreme flooding event from the Extreme Weather Event database.

3.3.5 Leads with no results

- **FENZ**: Fire and Emergency New Zealand was approached through Hamish McEwen and Craig MacAlpine to obtain operational records, similar to those collected in the South of France. However, no response was received.
- **NZ Police**: The New Zealand Police were contacted for the same purpose, through James Gunning, who informed us that no relevant data were available for sharing.
- **GEMA**: The Geographic Emergency Management Aotearoa (GEMA) is a GIS community that facilitates collaboration and coordination during disaster response efforts and played an important role during the Cyclone Gabrielle response in 2023. After discussions with members of this community (notably Kate Burns), it was confirmed that GEMA does not retain any operational datasets.
- **Private insurance companies**: As previously discussed, flood-related house damage is typically covered by private insurance companies. After discussions with Professor Ilan Noy (Chair in the Economics of Disasters and Climate Change), who has spent nearly a decade unsuccessfully attempting to obtain private insurance claim data, it was concluded that pursuing this lead would not be worthwhile.

4 Impact modelling

4.1 RiskScape

RiskScape [Paulik et al., 2023] is an open-source software developed by NIWA to assess the impacts of multiple hazards on buildings. In practical terms, it can integrate a variety of input layers and spatially combine them to produce impact estimates.

In our application, RiskScape was used as the impact component of the impact-based flood forecasting chain. It takes as input the layer of buildings representing the June 2021 situation in Westport (extracted from ©LINZ), combines it with the hazard layers (i.e., the flood maps produced for each forecast member), and calculates the flood depth each building is exposed to.

Comparison with the impact component used in France

In the flash flood nowcasting chain implemented in the Mediterranean region, a similar process is used: the water depth is calculated for each building. The method of calculation is also comparable: for each building polygon, the maximum water depth value among all intersecting flood map pixels is retained.

Based on the AI-based physical damage prediction model described in Paulik et al. [2024], RiskScape also provides a damage ratio (i.e., cost to repair divided by cost to replace) for each building in Westport. It is important to note, however, that this damage model was trained on the same flood event that we are reforecasting, although using a different observation dataset (see Section 3.3.1).

Water depth threshold for building damage

In the damage model implemented for Westport [Paulik et al., 2024], the damage ratio is zero if the flood depth is less than 15 cm. It is reassuring to note that this 15 cm threshold was also adopted in the PhD study for considering a building as damaged, corresponding approximately to a typical door threshold. However, in the PhD, the impact-based model was designed to issue warnings based on a threshold number of buildings affected within each spatial entity (river reach or municipality).

In concrete terms, RiskScape was applied in this study by using the following command (Figure 8):

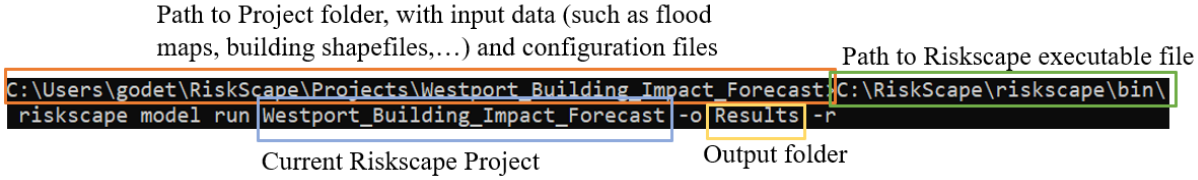


Figure 8: Riskscape command.

This system command can be directly integrated into the forecasting routine, allowing for the sequential calculation of flow, inundation, and impacts. Figure 9 shows the structure of the RiskScape project folder, which contains all necessary input data and configuration files. The folder was provided as-is for the purposes of this study and required minimal adaptation (e.g., updating file paths in ”2021_Westport_Ensemble.csv”, ”project.ini”, ”bookmarks.ini”, or in ”2021_Westport_Flood_Impacts_18032025.py”).

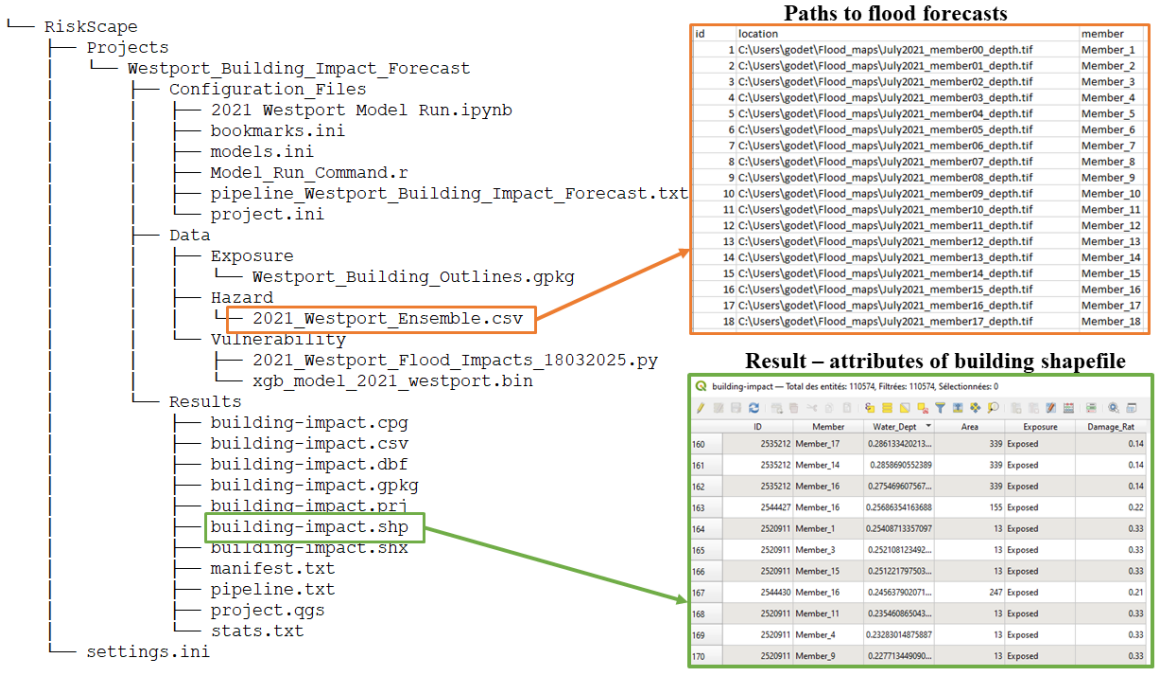


Figure 9: Riskscape folder.

Perspectives on the use of Riskscape in a flood forecasting routine

In the forecasting chain implemented in this study, RiskScape is used as the impact model, estimating the water depth to which each building is exposed. This represents a basic application of the software—an operation that could alternatively be performed through simple raster comparison, as was done in the French case study during the PhD. However, RiskScape has been optimised for fast execution and includes well-considered methods for determining which water depth value to assign to each building (e.g., pixel under centroid, mean or maximum of all pixel values within the building outlines). While its use in this routine does not introduce substantial added value from a methodological standpoint, it does save time and effort, leverages locally developed tools, and offers potential for broader applications.

4.2 Binary impact detection

The first step of impact-based modelling applied here focuses on building damage and produces a binary output: whether a building is affected or not. Based on the impact outputs provided by RiskScape, two approaches were considered:

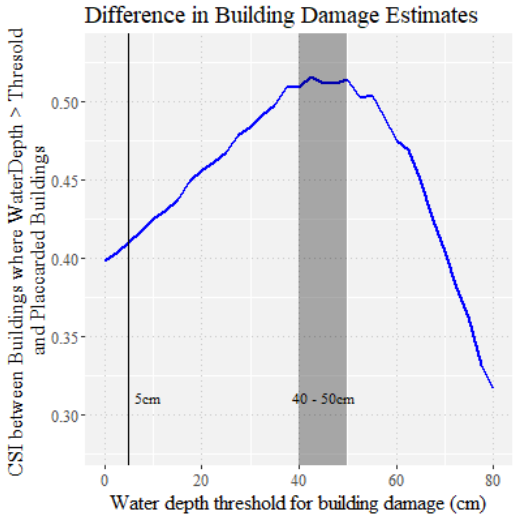
- Using the water depth each building is exposed to and applying a threshold on flood depth.
- Using the damage ratio and applying a threshold on the damage ratio value.

Comparison with the impact component used in France

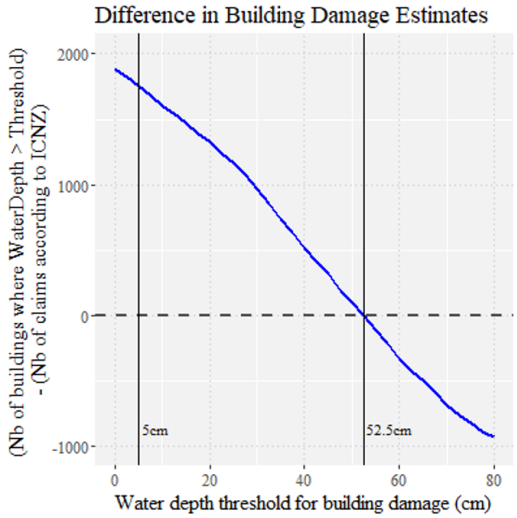
In the flash flood nowcasting chain implemented in the Mediterranean region, a water depth threshold of 15 cm was applied to each building. This value was chosen because it roughly corresponds to the typical height of a door threshold. However, no rigorous sensitivity analysis could be conducted at the time. In contrast, the Westport case study and the available datasets offer the opportunity to explore the choice of threshold more systematically.

To determine the most suitable parameter and associated threshold for binary impact forecasting, the ”observed” flood map presented in Section 2.4.3 and Figure 5 was used to compare the ”best” simulated impacts (i.e., impacts derived from the best flood map simulation) against observed impacts described in Sections 3.1 (insurance claims from ICNZ) and 3.2 (placard data). Specifically:

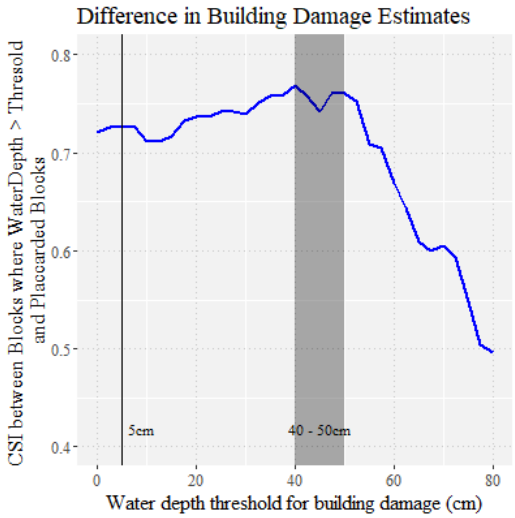
- A CSI score was calculated by comparing the buildings (or blocks) classified as affected according to the placard data with those classified as affected in the simulations, based on thresholds applied to water depth or damage ratio. Results for thresholds ranging from 0 to 80 cm (for water depth) and from 0 to 1 (for damage ratio) are presented in Figures 10a, 10c, and 10d.
- The difference between the number of buildings identified as affected in the simulations and the number of insurance claims reported by ICNZ was calculated. Results for water depth thresholds between 0 and 80 cm are shown in Figure 10b.



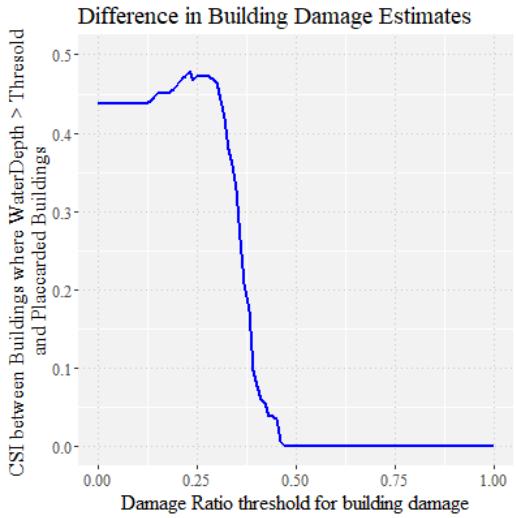
(a) Water threshold - Placard data - Building scale



(b) Water threshold - Number of claims from ICNZ - Building scale



(c) Water threshold - Placard data - Block scale



(d) Damage ratio - Placard data - Building scale

Figure 10: Sensitivty analysis to the selected parameters (i.e. water depth or damage ratio), based on selected validation data (i.e. placard data or number of claims from ICNZ), at the selected spatial scale (i.e. building scale or block scale).

From this sensitivity analysis, we retained the following conclusions:

- Using the damage ratio instead of the water depth does not lead to better results. Therefore, the water depth was selected as the parameter to consider a building as affected by flooding.
- Across both building and block scales, and for both observation datasets (placard data and insurance claims from ICNZ), a water depth threshold of 50 cm appears to offer the best compromise between over- and under-detection.
- Even when using the "best" flood map, the CSI score does not exceed 0.55 at the building scale and 0.75 at the block scale. Consequently, the validation scores obtained with the flood forecasting chain are expected to be limited by these values. Several explanations can be advanced:
  - Data and model limitations: The "best" flood map remains an approximation rather than a true observed flood map; observed impacts may not perfectly reflect actual ground truth; and the impact model based on a simple water depth threshold inevitably simplifies the complexity of flood damage processes.
  - Findings from other studies: Previous research has shown that a significant proportion of flood damage is diffuse, unexplained by flood modelling alone, and linked to complex runoff processes (see below).

A French feedback on low impact detection scores

- Le Bihan et al. [2017] compared insurance claims related to two flash flood events in the Mediterranean region to different flood scenarios. They found that a significant proportion of claims were located outside the estimated 1000-year flood envelope. They suggested that these claims could have been caused by unmodelled flood mechanisms such as runoff accumulation in low points, sewer saturation, or cellar flooding due to rising groundwater levels.
- Moncoulon et al. [2020] performed a broader analysis by comparing all flood claims recorded in the Caisse Centrale de Réassurance database to flood hazard maps for the French Mediterranean region. These maps accounted for both riverine and surface flooding. However, they observed that almost 40% of reported damages were located outside all mapped flood-prone areas, even when considering runoff modelling in upstream areas. They noted that runoff-related damage does not follow hydrographic patterns and that, while it is driven by water height and speed, its paths are much more diffuse and unpredictable than those of river floods. Improved knowledge of drainage and stormwater evacuation network operations could enhance understanding of exposure to runoff.

4.3 Multi-category damage detection

The placard dataset, which includes information on observed damage severity, offers an opportunity to assess whether flood forecasts can go beyond binary outputs (affected / unaffected) and provide insight into damage severity. This section investigates whether such multi-category damage detection can be achieved without significantly compromising forecast accuracy.

4.3.1 At the building scale

To enable multi-category damage detection at the building scale, three water depth thresholds (T_1, T_2, T_3) are defined to match the damage categories present in the placard dataset:

- **Unaffected:** $0\,\text{cm} \leq WD < T_1$
- **Light damage:** $T_1 \leq WD < T_2$

- **Moderate damage:** $T_2 \leq WD < T_3$
- **Heavy damage:** $T_3 \leq WD$

Based on the results from Section 4.2, T_1 is set to 50 cm. The remaining thresholds T_2 and T_3 are determined through a sensitivity analysis using the reference flood map and the placard data at the building scale. For each tested threshold combination, a CSI score is computed for each category, and the values that maximize the total CSI sum are retained. The final thresholds are:

- **Unaffected:** $0\,\text{cm} \leq WD < 50\,\text{cm}$
- **Light damage:** $50\,\text{cm} \leq WD < 65\,\text{cm}$
- **Moderate damage:** $65\,\text{cm} \leq WD < 110\,\text{cm}$
- **Heavy damage:** $110\,\text{cm} \leq WD$

4.3.2 At the block scale

As explained in Section 3.2 and illustrated in Figure 6, aggregating damage severity raises methodological questions. A straightforward option is to retain the maximum damage category observed within each block. However, this approach does not consider the number of damaged assets within the block.

As an alternative, we define a continuous *Impact Score* by associating each damage category with a numeric value, as shown in Table 3:

$$\text{DamageNb} = \begin{cases} 0 & \text{Unaffected} \\ 1 & \text{Light damage} \\ 2 & \text{Moderate damage} \\ 3 & \text{Heavy damage} \end{cases}$$

For a block containing N buildings, the aggregated score is calculated as:

$$\text{ImpactScore}_{\text{block}} = \sum_{i=1}^{N} \text{DamageNb}_i$$

This approach results in a continuous variable, which takes values between 0 and 87 when applied to the observed placard dataset.

Similarities with the French Study Case

This approach is consistent with the impact simulations used to compare modelled impacts with the rankings provided by the Caisse Centrale de Réassurance (CCR). Specifically, CCR supplied ranked lists of impacted areas based on the number of insurance claims for numerous flood events in the French Mediterranean region. Corresponding simulated rankings were produced by ordering spatial entities according to the number of buildings affected in the simulations (i.e., buildings exposed to water depths exceeding 15 cm).

5 Cascading uncertainties

As illustrated by the structure of this report (Section 2), the impact-based flood forecasting chain consists of a sequence of interconnected modelling components. While evaluating the performance of each component independently is important, it is essential to also consider the forecasting system as a whole. A comprehensive understanding of how uncertainties propagate through the chain — and how the different components interact — remains a key challenge in the literature, particularly in the context of impact-based forecasts.

Titley et al. [2024] analysed the relationships between track, precipitation, and river discharge skill within a flood forecast chain, for one flood event (Hurricane Iota, which hit Central America in November 2020). Building on this, the present study extends the analysis by incorporating the flood mapping and impact modelling components, offering a broader view of the cascading uncertainties throughout the entire forecast chain.

Expanding the scope of forecast evaluation in the PhD

In the PhD work focusing on the French Mediterranean region, each component of the flood forecasting chain was rigorously evaluated across a wide spatio-temporal scale — including continuous hydrological validation over more than 10 years for the flow component, comparisons between observed and simulated flow–stage relationships at nearly 200 hydrometric stations for the flood library component, and continuous comparison of simulated warnings to insurance claim data over more than 10 years for the impact component. However, the propagation of uncertainty through the forecasting chain was never explicitly analysed at the event scale. The New Zealand case study offers an opportunity to explore this dimension in greater detail.

5.1 Forecast performance through the chain

Using the forecast and validation data described in the previous sections, errors were calculated for each block of the flood forecasting chain. Boxplots representing the ensemble spread for each forecast cycle are presented in Figure 11. Specifically:

- For rainfall forecasts, the spatially averaged catchment rainfall over the preceding 48 hours was calculated at each time step. The maximum value was then compared to the best available proxy for 48-hour maximum rainfall, and the difference between the two was computed. Results are shown in Figure 11a. Additional indicators were also tested, including 24-hour maximum rainfall, total accumulated rainfall, and Nash-Sutcliffe Efficiency (NSE) scores for 3-hour and 6-hour rainfall accumulations. These results are presented in Figure 17.
- For flow forecasts, the Kling-Gupta Efficiency (KGE) was computed at each time step for the forecasted flows at the Te Kuha station, and compared to the KGE calculated for the best proxy flow. These errors are illustrated in Figure 11b. In addition, Figure 17 presents results based on an alternative performance metric: the difference in peak flow values.
- For flood forecasts, forecasted flood water levels were compared to observed water levels (based on the flood map in Figure 5), using the Root Mean Square Error (RMSE) computed over all 5 m $\times$ 5 m pixels. Results are shown in Figure 11c.
- For impact forecasts, affected blocks identified in the placard dataset were compared to simulated affected blocks (defined as blocks with at least one building exposed to water depths exceeding 50 cm). This binary comparison enabled the calculation of the Critical Success Index (CSI), with results presented in Figure 13d. Additional CSI results at the building scale are presented in Figure 13a, and boxplots comparing model outputs to the total number of claims reported by ICNZ are shown in Figure 13c.

For presentation purposes, all boxplots show absolute errors, and CSI (resp. KGE) values are converted to $1 - \text{CSI}$ (resp. $1 - \text{KGE}$) to allow for a consistent visual interpretation of decreasing error trends.

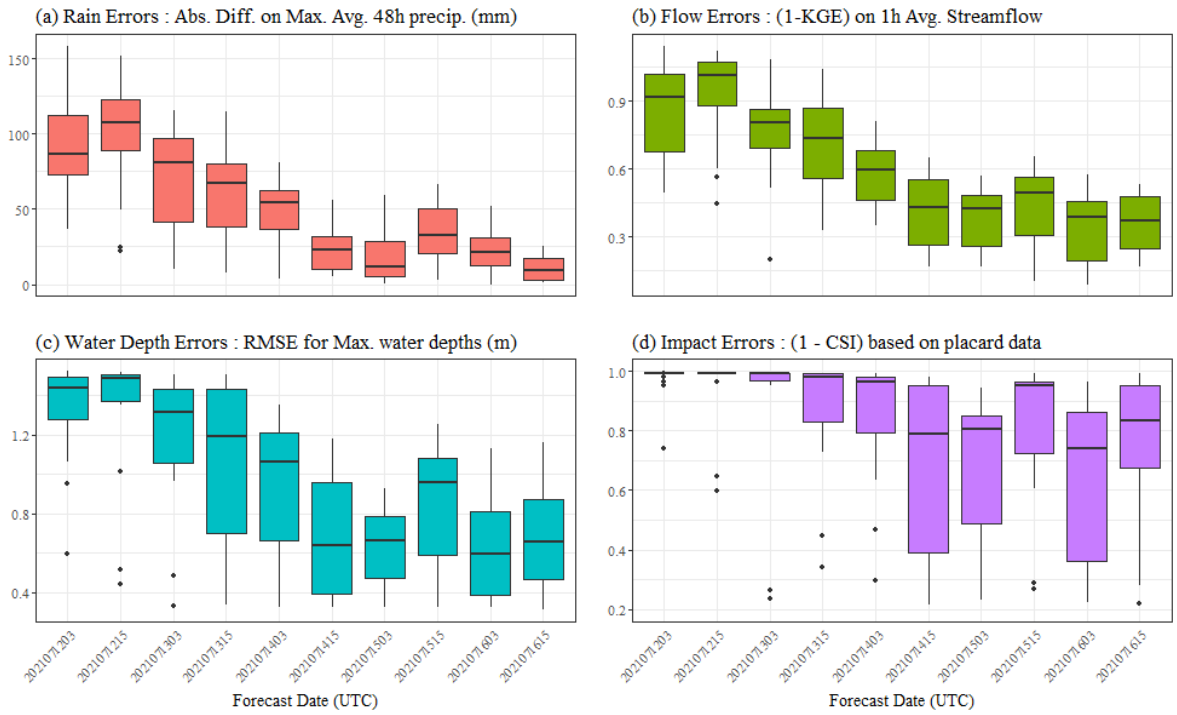


Figure 11: Boxplots comparing forecast performance across the ensemble members for each forecast cycle

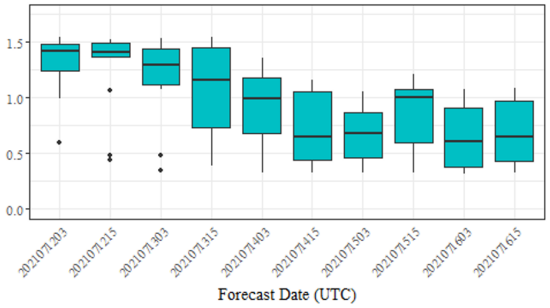
Below are some comments on these results:

- The rainfall forecasts consistently improve up until 2021071415UTC, approximately three days before the event. An unexpected drop in skill is then observed in the 2021071503UTC forecast cycle, leading to a performance degradation that propagates through the entire forecasting chain. Between the final two forecast runs (2021071603 and 2021071615), the rainfall skill improves again; however, this improvement does not translate across the rest of the chain. The skill between the last two forecasts does not change much for the flow and flood components (this is confirmed in figure 14), but it significantly deteriorates at the impact level. This non-linear response highlights the complex propagation of uncertainty and the sensitivity of downstream components to modest upstream changes.

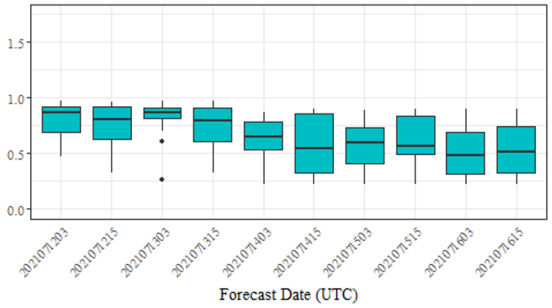
- From 2021071415UTC onward, the performance of streamflow forecasts stabilises, with the median KGE converging toward 0.6 $(1 - \text{KGE}) \approx 0.4$, a value generally considered indicative of "good" predictive skill. However, the event was still markedly underestimated, as shown by the hydrographs in Figure 3 and the correlation plots in Figure 17, with most rainfall and discharge errors (Sim. – Obs.) being negative.

- Apart from the skill degradation observed at 2021071515UTC—triggered by a deterioration in rainfall forecast skill—the median RMSE for water depths remains around 60 cm in the last forecast cycles. This error must be interpreted in light of the expected uncertainty in the "observed" water depths used for comparison, which can reach ± 30–50 cm according to Bates [2022]. Moreover, the initial RMSE values were computed over the entire simulation domain, which includes areas far beyond the urban perimeter of Westport (see purple rectangle in Figure 12c). When the validation domain is restricted to the impacted area only (see red rectangle), the RMSE improves, dropping closer to 50 cm

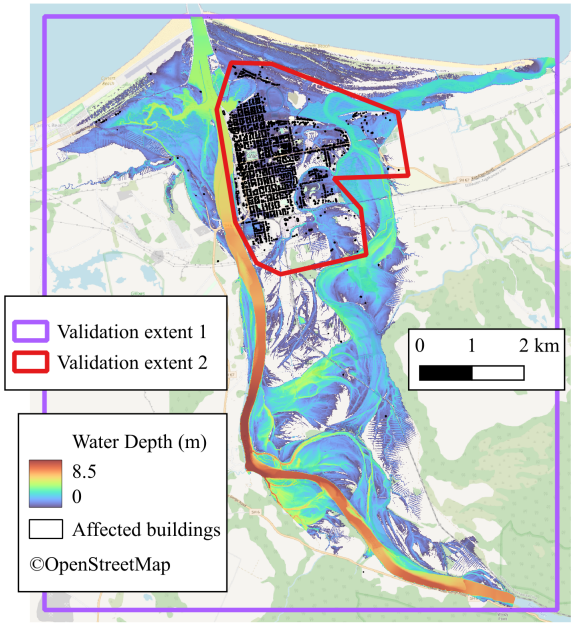
in the final forecasts (see figure 12b). Nonetheless, even a 50 cm median error significantly affects the impact model's performance, especially given that the threshold for a building to be considered "affected" in the simulation is also set at 50 cm, as detailed in Section 4.2.



(a) RMSE (m) from validation extent 1



(b) RMSE (m) from validation extent 2



(c) Extents on which the flood validation was performed

Figure 12: Comparison between simulated and observed maximum water depths during the event, evaluated over two different flood extents.

- At the impact level, the skill (translated by the CSI score) is limited and does not exceed 0.8 for the best-performing member, as explained in section 4.2. Even for the best performing forecast cycle, the median CSI is low (0.26).

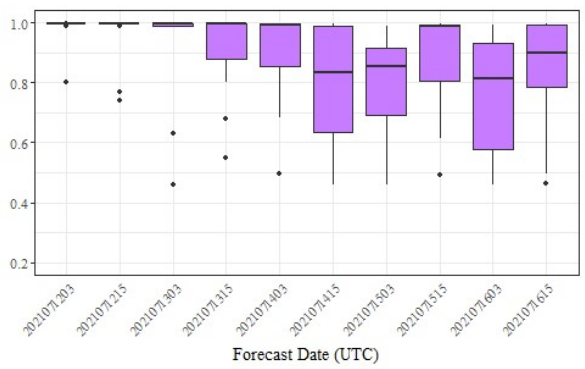
Comparison with impact scores obtained in France

Some explanations for the low impact detection scores have been provided in Section 4.2. Additional insights can be drawn by comparing these results to those obtained from the French case study. Over the last five forecasts, the median CSI values for the impact component at the block scale in the Westport case study (Figure 11d) were 0.21, 0.20, 0.05, 0.26, and 0.17, respectively, yielding an average of 0.18.

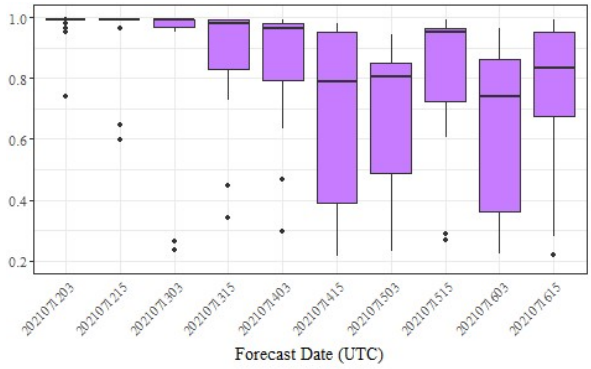
In France, for the continuous validation of the impact-based nowcasting chain at the river reach scale—using insurance claims as the reference—the obtained CSI was 0.17.

Although the differences in validation window (event-scale vs. continuous), spatial

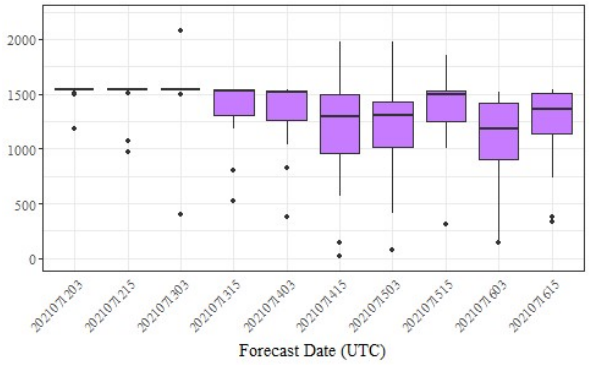
scale (0.03 km^2 for blocks vs. 4 km^2 for river reaches), domain (city scale vs. regional scale), and forecast range (multi-day forecasts vs. nowcasts) make direct comparison difficult, the structure of the forecasting chains and the validation method are broadly similar. Despite these differences, the similarity in CSI values is noteworthy.



(a) CSI based on placard data at the building scale



(b) CSI based on placard data at the block scale



(c) Abs. Diff based on ICNZ number of claims

Figure 13: Boxplots of impact errors using different validation datasets

5.2 Run-to-run jumps in forecast performance

As in Titley et al. [2024], we calculate the divergence score as defined by Richardson et al. [2020]:

$$\text{Divergence} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}|x_i - y_j| - \frac{1}{2M^2}\sum_{i=1}^{M}\sum_{j=1}^{M}|x_i - x_j| - \frac{1}{2N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}|y_i - y_j|$$

where x_i and y_j are the error values for ensemble members i and j in two successive forecast cycles x and y, with $M = N = 18$ the number of members in each forecast.

This divergence score quantifies the change in ensemble forecast performance from one run to the next. By comparing divergence scores across the rainfall, flow, flood, and impact components of the forecasting chain, the aim is to assess how closely aligned the run-to-run shifts in performance are across the different components.

The results, presented in figure 14, show that the largest change across all variables occurs between the 2021071415 and 2021071503 runs, with a particularly pronounced peak in rainfall divergence. This marks a clear transition in the forecast system, where the rainfall ensemble undergoes both a substantial shift in median values and a reduction in spread (Fig. 13). The matching peaks in flow, water depth, and impact divergence indicate that this change in rainfall forecast propagated through the entire forecasting chain.

By contrast, other forecast transitions such as 2021071515 to 2021071603 show low divergence for rainfall, but a more marked divergence for water depth and impact forecasts. This suggests that even when the rainfall forecast remains structurally stable, changes in downstream components of the forecasting chain can still occur.

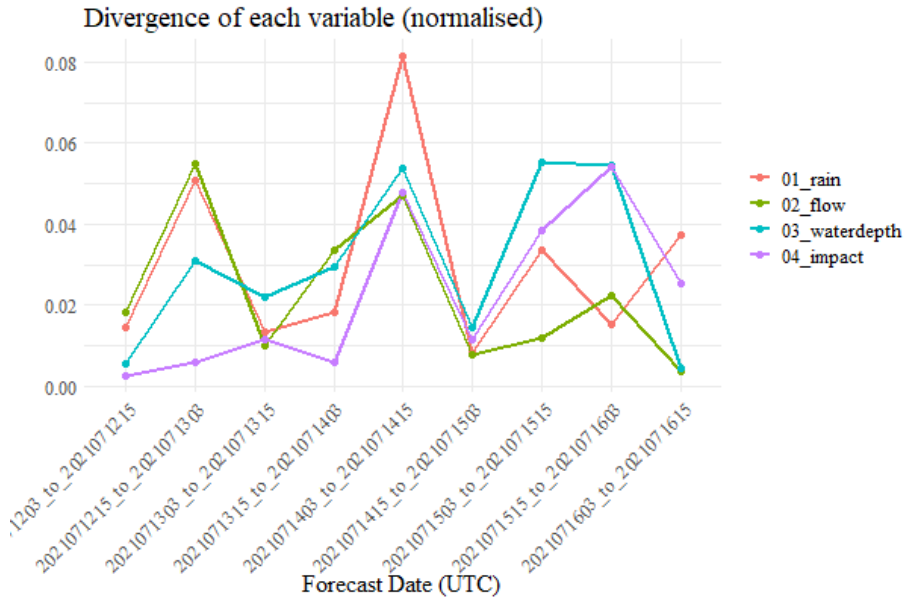


Figure 14: Divergence score of all error metrics, normalized by the y-axis range

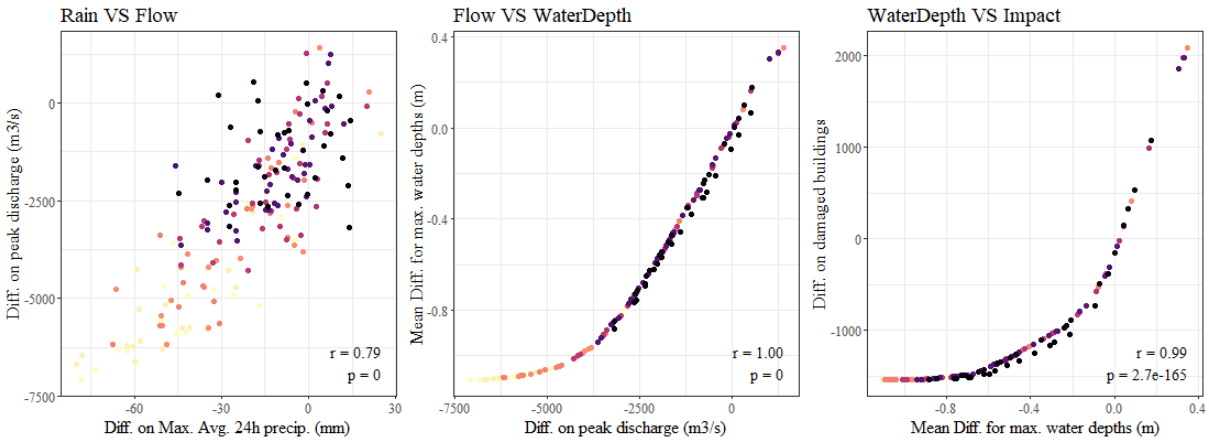
5.3 Correlation of Errors

Another way to represent the cascading uncertainties throughout the flood forecasting chain is through error correlation graphs, as proposed by Titley et al. [2024]. Rainfall–flow error indicators are shown in Figure 17, while correlations between flow and inundation, and between inundation and impacts, are presented in Figure 15. The results indicate that:

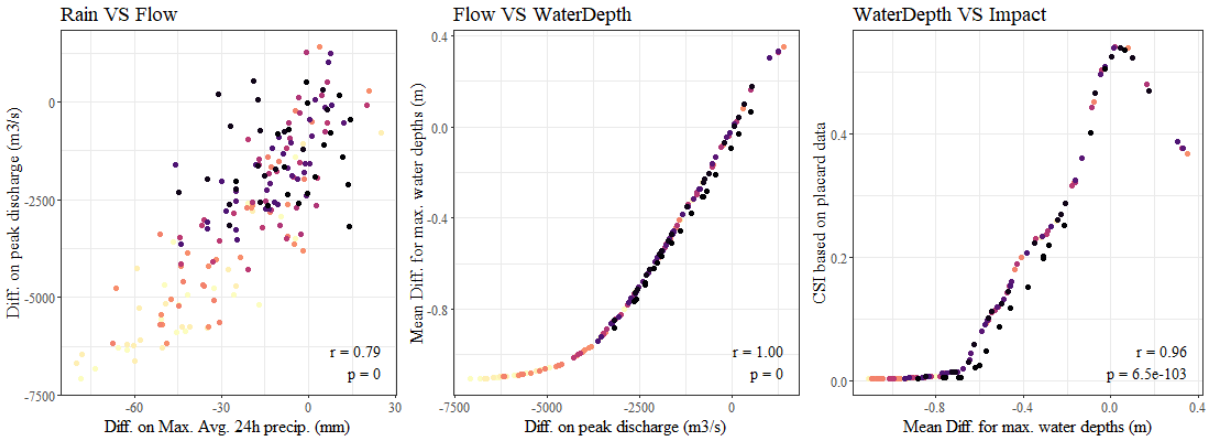
- There are very strong correlations between flow and inundation errors, and between inundation and impact errors, indicating tight coupling across components of the modelling chain.
- In particular, the perfect correlation between peak discharge error and water depth error (Spearman $r = 1$) suggests that the 2021 Westport flood was primarily driven by peak flows, with limited influence from flood volume or tidal/sea-level components.

> **Implications for Hydraulic Modelling**
> Compared to the Mediterranean case study, these results imply that a steady-state hydraulic modelling approach, as used operationally in France, would likely have provided satisfactory results for this specific flood event.

  The correlation appears linear over a wide range of flow errors ($[-2500, 0]$ m^3/s), but the present dataset does not allow investigation of model behaviour under flow overestimation scenarios.
- The correlation between water depth and impact prediction performance is also strong. The Spearman correlation between water depth error and impact error is slightly lower when validation is performed at the building scale using placard data ($r = 0.96$, Figure 15b), compared to event-scale validation using total claims from the ICNZ dataset



(a) Errors in each model component: Rainfall error is the difference (Sim - Obs) in the temporal maximum of spatially averaged 24-hour catchment precipitation. Flow error is the difference in peak discharge. Inundation error is the mean difference in maximum water depths. Impact error corresponds to the difference in the number of affected buildings, using the ICNZ total claims as a reference.



(b) Same as (a), except for the impact component, which is validated at the building scale using placard data and the CSI as the evaluation metric.

Figure 15: Scatterplots of errors in each model component; each dot represents one ensemble member. Members are colored by forecast run (darker = more recent). Spearman correlation coefficients (r) and p-values are reported in each panel.

($r = 0.99$, Figure 15a). This small difference likely reflects the increased complexity and noise introduced at finer spatial scales, where model inaccuracies, local vulnerability factors, and observational uncertainties become more prominent.

Figure 15b shows that achieving a Critical Success Index (CSI) above 0.5 for the prediction of impacted buildings typically requires the mean water depth error to remain within approximately $[-0.05\,\text{m}, 0.1\,\text{m}]$. This range is slightly asymmetric, potentially indicating greater tolerance for overestimation than underestimation. However, due to the limited number of ensemble members achieving high CSI values, further analysis would be needed to confirm this pattern.

To investigate these relationships at the building scale, Figure 16 shows the distribution of HIT, MISS, and FALSE ALARM classifications based on the last five forecast cycles across all ensemble members. This complements the domain-scale analysis by highlighting the variability of impact predictions at individual building locations:

- The FALSE ALARM distribution peaks near zero error, underscoring the sensitivity of threshold-based impact models: even small overestimations around the 50cm threshold can lead to false positive predictions.

- The MISS distribution peaks around -0.5m, indicating that in many false negative cases, the model underestimated water depth by about 50cm. A secondary peak near 0m suggests that even minor underestimations can lead to missed impacts if they occur near the threshold.
- Unlike Figure 15b, which aggregates errors across the domain, this figure shows that all three outcomes (HIT, MISS, FALSE ALARM) can still occur at near-zero water depth error. This highlights the influence of local uncertainties and the limitations of binary classification. Nonetheless, the peak of the HIT distribution at 0m confirms that accurate water depth predictions generally yield correct impact classification.

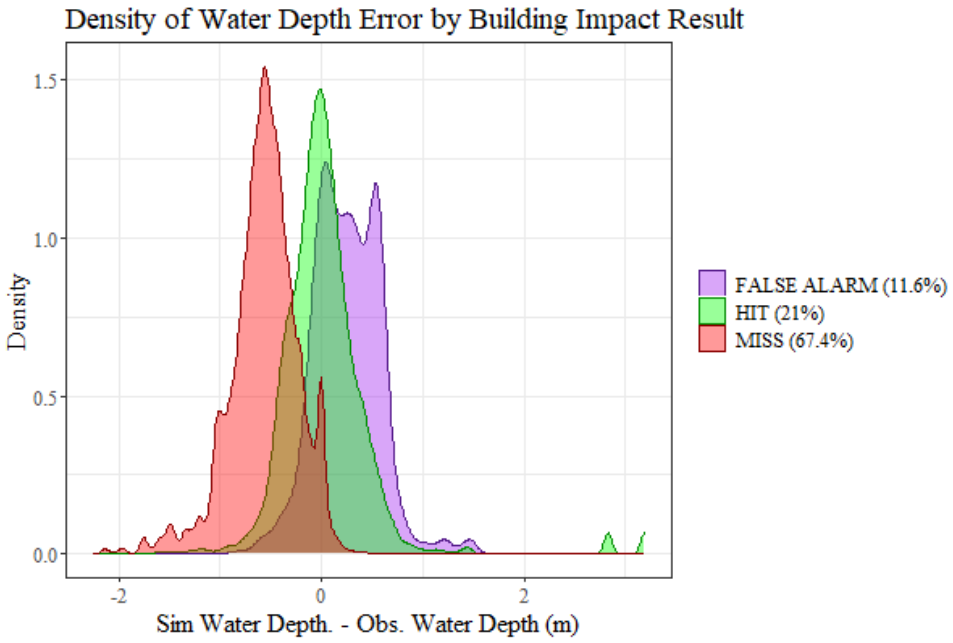
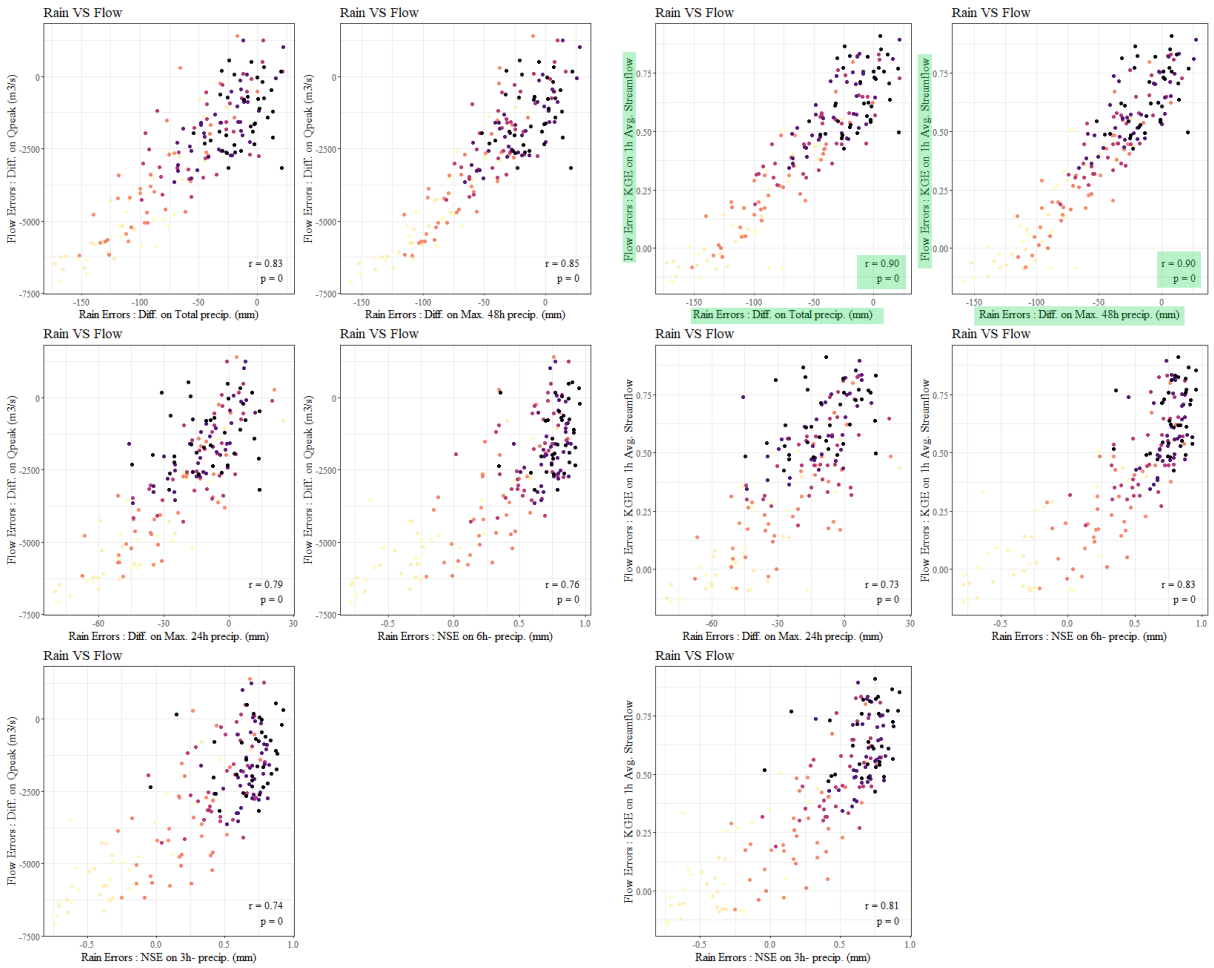


Figure 16: Density of Water Depth Error by Building Impact Result. The x-axis shows the water depth error at each building location, calculated over the last five forecast cycles and across all ensemble members. Each building is assigned a contingency category (HIT, MISS, or FALSE ALARM) based on whether the simulated water depth exceeds 50cm, and whether the building was recorded as damaged in the placard dataset.

- As shown in Figure 17, the strongest correlation between rainfall and streamflow occurs when the rainfall error is defined as the difference in total or maximum 48-hour catchment precipitation, and the streamflow performance is measured using KGE. This highlights the importance of accurate cumulative rainfall estimation for runoff modelling. In contrast, rainfall metrics based on short-term dynamics (e.g. 3-hour or 6-hour NSE) are less strongly correlated with flow error, suggesting that runoff generation in this case is more sensitive to total rainfall volume than to sub-daily temporal distribution. Furthermore, the higher correlations obtained with KGE (compared to peak discharge error) confirm that rainfall accuracy influences the full hydrograph shape, not just flood peak values.



(a) Rainfall–flow error correlations using peak discharge as the flow metric, and various rainfall indicators.

(b) Rainfall–flow error correlations using KGE as the streamflow metric, and various rainfall indicators.

Figure 17: Comparison of rainfall–flow error correlations using different metrics for both rainfall and flow components.

6 Comparing quantile-based approaches

Until now, we have propagated the information through the different components of the forecasting chain using individual ensemble members. However, another possibility is to use probability information such as minimum, maximum, or quantiles. With this second method, it is interesting to study how the final result (i.e., the impact forecast) varies when the probability is calculated at varies locations of the forecasting chain. To illustrate this, let's schematize the ensemble flood forecasting chain that simulates successively the discharge at Te Kuha (Q), the inundation on the simulation domain ($Inundation$) and the impacts in terms of building damage ($Impact$). The quantiles of the ensemble (i.e. 50% quantile, 90% quantile...) can be computed at different locations:

$$\begin{cases} Impact(Inundation(\textit{quantile}(Q))) \\ Impact(\textit{quantile}(Inundation(Q))) \\ \textit{quantile}(Impact(Inundation(Q))) \end{cases}$$

It is interesting to assess how the impact forecast varies depending on how and where the probabilities are calculated, as it helps understanding how probability information and error propagate through the cascade of forecasts. In order to do that, we have considered a lagged-ensemble using the 5 last forecast cycles, to be able to build a $18 \times 5 = 90$ members ensemble which is of more appropriate size to calculate quantile values.

The results, in figure 18 show very few differences in the final results for the considered probabilities: 50%, 75%, 85%, 90%, 95% and max. This indicates that for this event, the forecasting chain is quite stable and monotonous: high flow values will result into high water depths which in turn yield significant impact. Then it also means that in this case, the simplest approach: $Impact(Inundation(\textit{quantile}(Q)))$ would be sufficient. In operational terms, this is quite important since it would siginficantly reduce computation times.

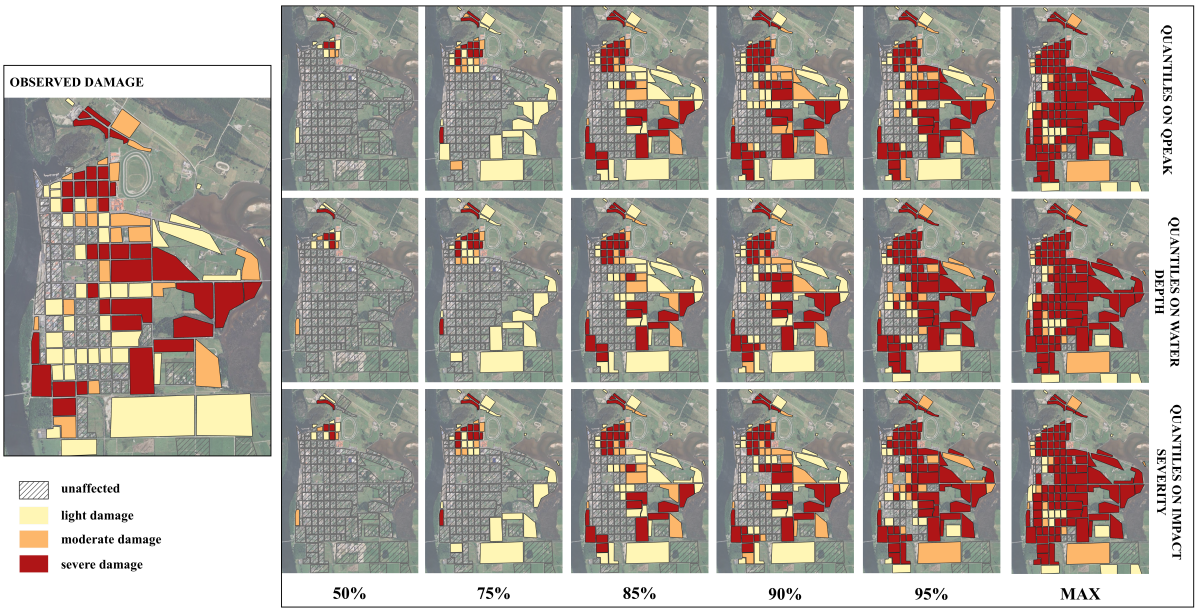


Figure 18: Impact (damage severity aggregated at the block scale) observed based on the placard dataset (left), and forecasted for different probabilities, with probabilities calculated on peak flows at Te Kuha (top), on the water depth at the pixel-scale (middle), and on the ranking of affected blocks (bottom).

7 Flood library and interpolation methods

In order to assess the impact of flood library selection and interpolation method on forecast performance—and to ensure comparability with the French case studies—the following approaches were tested:

- **Flood library selection (15 scenarios):**
  - *Uniform library:* Scenarios are selected from the full set of $\simeq$100 simulations using a maximum dissimilarity algorithm to ensure uniform coverage of the 2D parameter space defined by standardised peak sea level and standardised peak flow.
  - *Design library:* Scenarios are selected to represent typical fluvial and coastal return periods, initial fluvial climate change simulations (limited RCPs), sea level rise increments, additional RCP-based fluvial scenarios, and frequent low-impact coastal events.
- **Interpolation methods:**
  - *1D interpolation:* Linear interpolation within the 1D parameter space defined by peak 2-hour flow.
  - *1D nearest neighbour:* Selection of the scenario with the closest peak 2-hour flow, i.e., no interpolation.

Since interpolation is applied in a 1D space defined by flow at Te Kuha, only scenarios with distinct flow values will result in a distinct flood map. For each library, these flow values are presented in table 5.

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Design Library	7451	8514	9304	9493	9818	10564	10383	10738
Uniform Library	8514	11582	8175	3753	11944	5252	5945	5399

	9	**10**	**11**	**12**	**13**	**14**	**15**	**Obs.**
Design Library	11581	1000	1000	1000	1000	1000	1000	**8632**
Uniform Library	117	8994	8169	3096	1000	1000	12492	

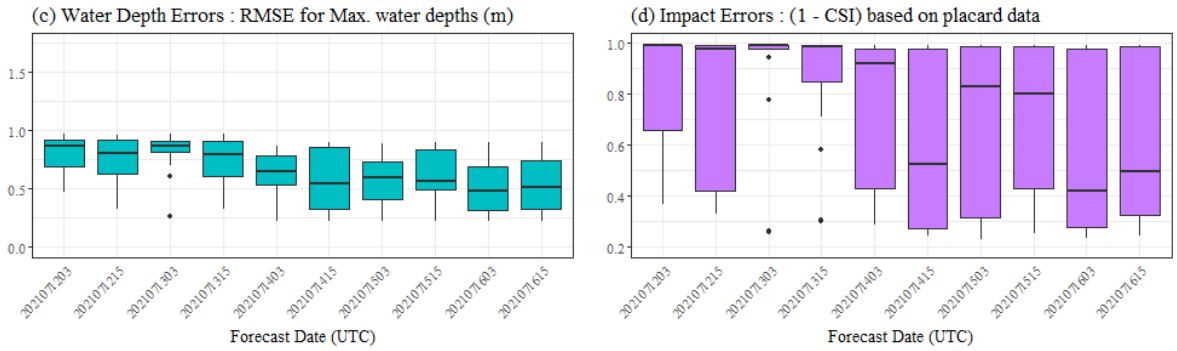
Table 5: Flow values at Te Kuha (m^3/s) for the 15 scenarios of each flood library considered.

The performance of the Inundation and Impact components for each configuration is illustrated with the boxplots presented in figure 19.

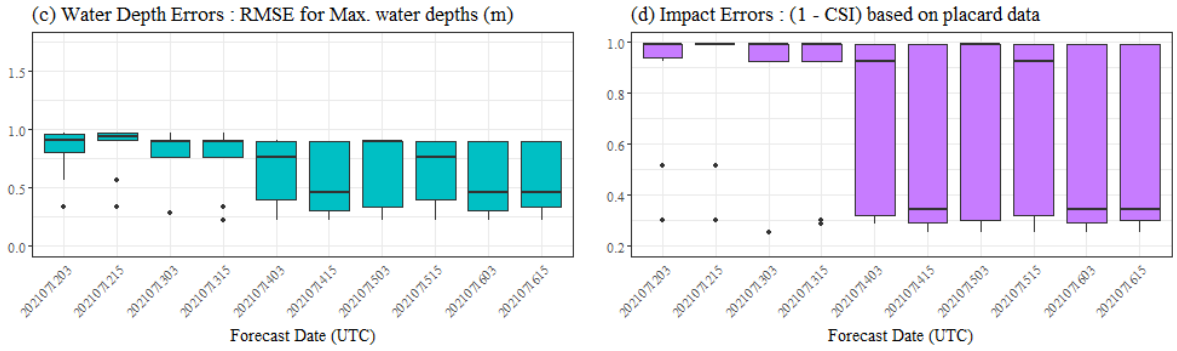
The results show a significantly larger spread and higher errors for the Uniform library. As seen in Table 5, this can be attributed to the greater representation of lower flow values within the Uniform library—seven scenarios have flow values below $8000\ \mathrm{m}^3/\mathrm{s}$, compared to only two in the Design library. Since the forecasts substantially underestimated the peak flow (see Figure 3a) and rarely reached the reference value of $8632\ \mathrm{m}^3/\mathrm{s}$, the Uniform library provided more low-flow scenarios, which ultimately led to larger errors.

In this case, the Design library performs better, but this appears to be mostly coincidental. Had the event been less extreme, the Design library would likely have poorly captured it.

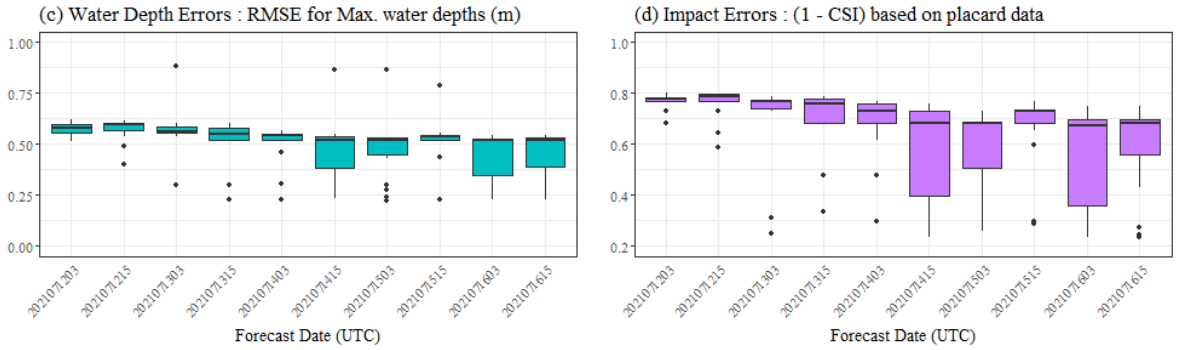
Regarding interpolation methods, linear interpolation generally yields slightly better or comparable results to nearest-neighbor, especially for inundation errors. However, in some cases—such as early forecasts using the Design library—nearest-neighbor interpolation performs similarly or even better.



(a) Uniform library, 1D linear interpolation method



(b) Uniform library, 1D nearest neighbor method



(c) Design library, 1D linear interpolation method

(d) Design library, 1D nearest neighbor method

Figure 19: Comparison of the performance of the Inundation and Impact component for the different flood libraries and interpolation methods tested.

8 Representing impact-based flood forecasts: user insights

Another important aspect of ensemble impact-based flood forecasts is their communication. The communication of ensemble forecasts is a well-known challenge and has been increasingly addressed in recent years. However, most of the existing literature focuses on rainfall and/or flow forecasts. The first study to examine more downstream components of the forecasting chain, specifically inundation forecasts, is Jean et al. [2023]. Key findings from that study are summarised below. Four prototype visualisations of ensemble inundation forecasts were presented to respondents from various user groups:

- Visualisation tools should be adapted to the specific needs of users.
- The amount of information displayed in a single graph should be limited.
- Users interpret information with inherent cognitive biases.
- Probability intervals are generally recommended for communication.
- Across all interview groups, forecasts were mainly considered useful for anticipating potential consequences by comparing to familiar reference events.
- Including both hydrographs and reference thresholds was seen as beneficial.
- Prototype 2—which presents low, median, and high scenarios without explicit probability values—was the most preferred.
- Prototype 4—which asks users to input a threshold and then shows the probability of exceedance—was not well received.
- The concept of probability remains widely misunderstood.

To the best of our knowledge, no study has yet focused on the communication of ensemble impact forecasts, which introduces additional specific challenges. To explore this gap, a survey was designed to investigate preferences regarding the content and visualisation of impact forecasts. It was distributed during a Webinar organised by the Resilient River Community (available at https://www.resilientrivers.nz/webinar-recordings). Two poll questions were displayed live during the Webinar (37 responses), and a more detailed survey was provided afterward (24 responses). Each survey question and the corresponding responses are presented below.

8.1 Questions and answers from the survey

8.1.1 Profile of respondents

Survey on flood warnings

This survey aims to better understand content and visualization needs for ensemble impact-based flood warnings. Your feedback is invaluable.
The estimated time to complete the survey is **5–10 minutes**. It contains **8 questions**.

* Indique une question obligatoire

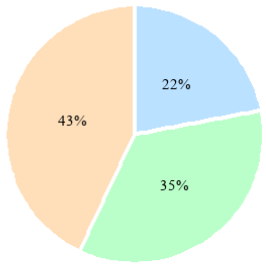
About you – In the context of flood warnings, which role best describes your primary position? *

- ◯ Member of the public (information receiver)
- ◯ Emergency manager or stakeholder (information receiver)
- ◯ Flood forecaster or flood expert (information provider)

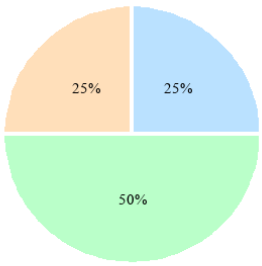
(Optional) Please specify your job title or role:

Votre réponse

(a) Poll responses



(b) Survey responses



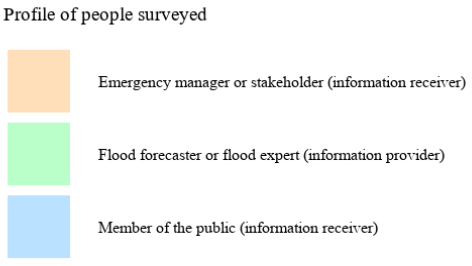
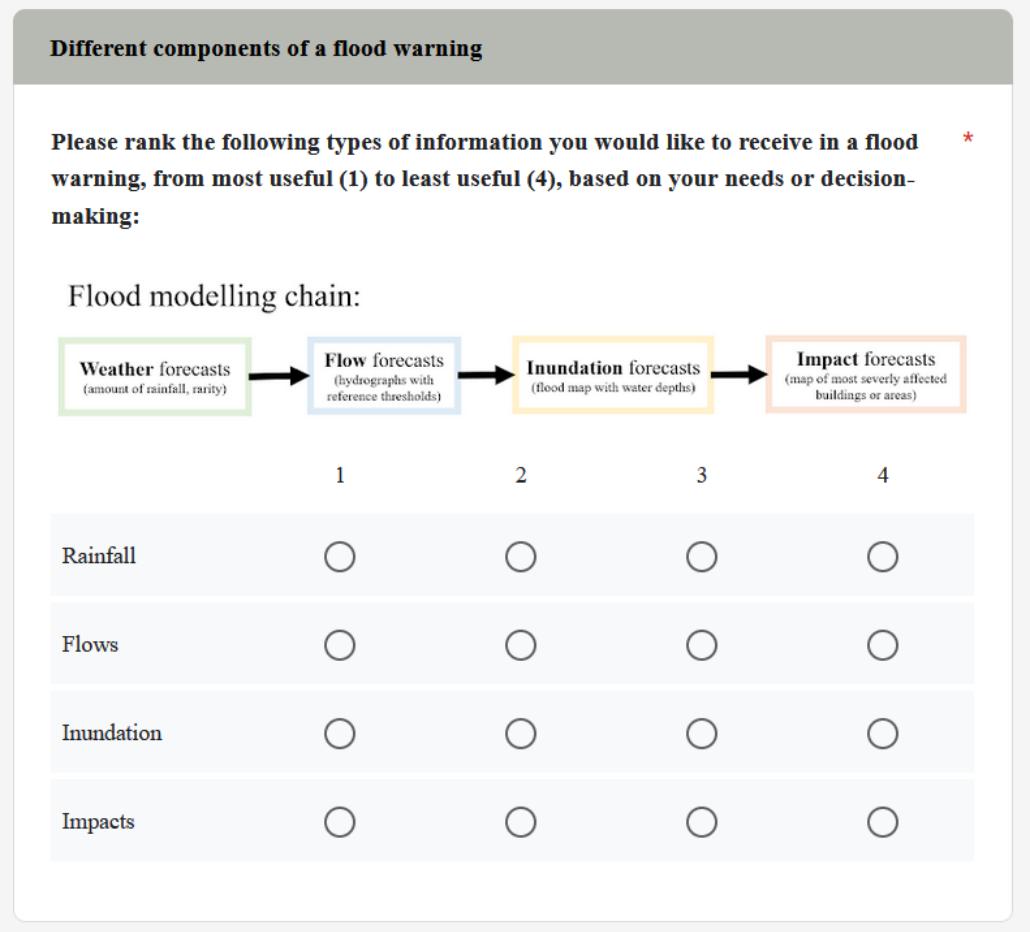


Figure 20: First question of the survey and results obtained - profile of respondents

8.1.2 Different components of a flood warning



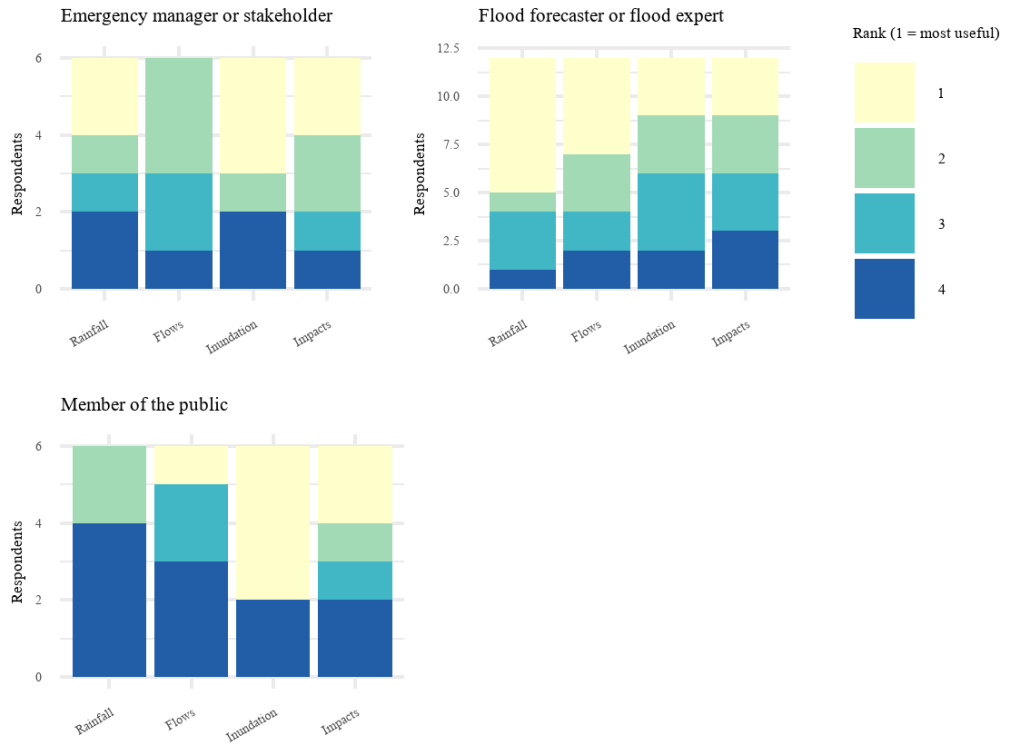
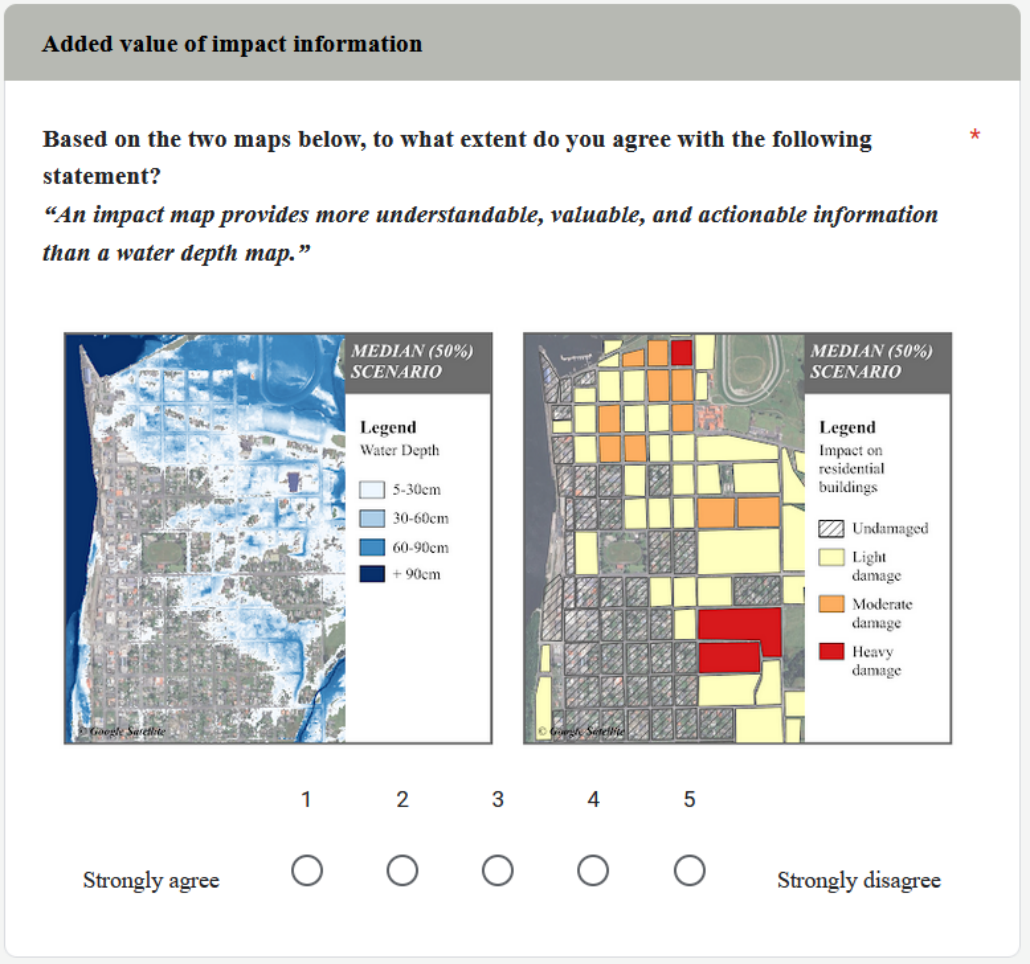


Figure 21: Second question of the survey and results obtained - different components of a flood warning

8.1.3 Added value of impact information



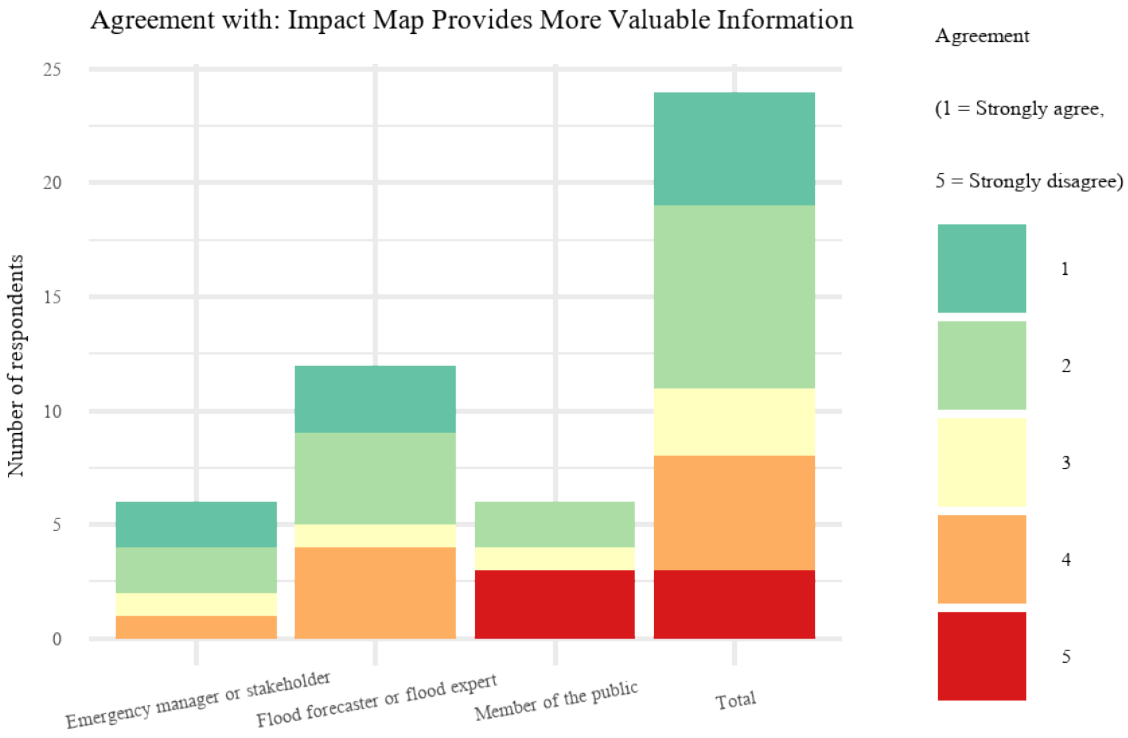


Figure 22: Third question of the survey and results obtained - added value of impact information

8.1.4 Trade-off between accuracy and familiarity

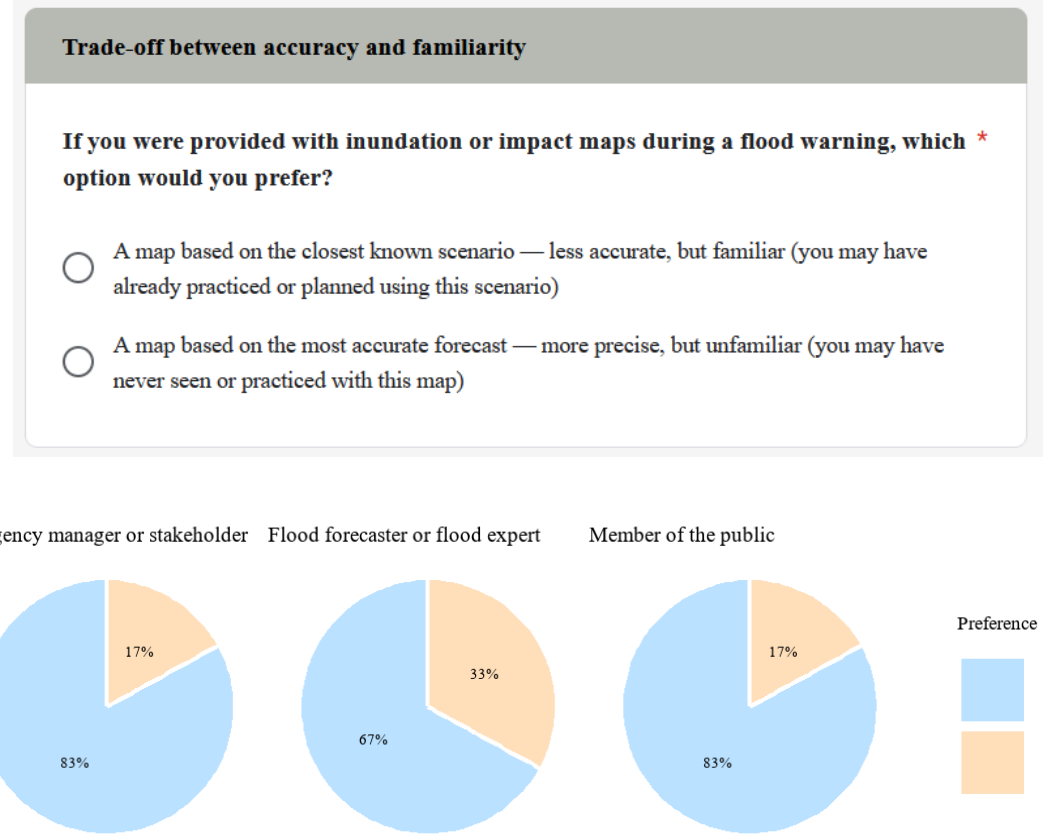
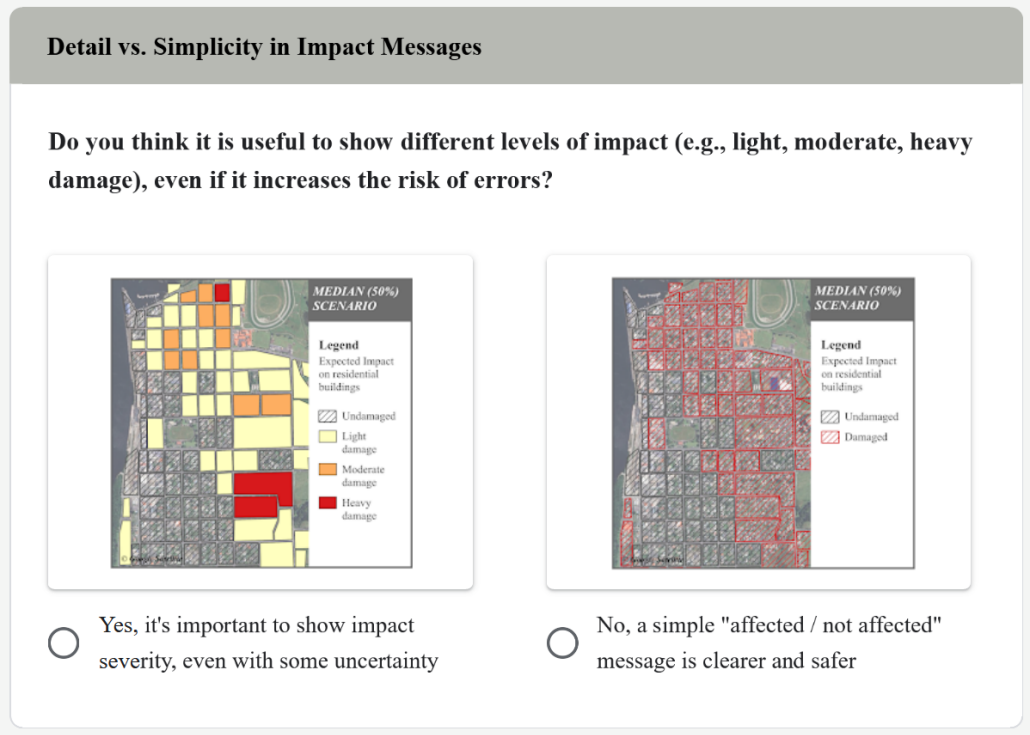


Figure 23: Fourth question of the survey and results obtained - trade-off between accuracy and familiarity

8.1.5 Detail vs. Simplicity in Impact Messages



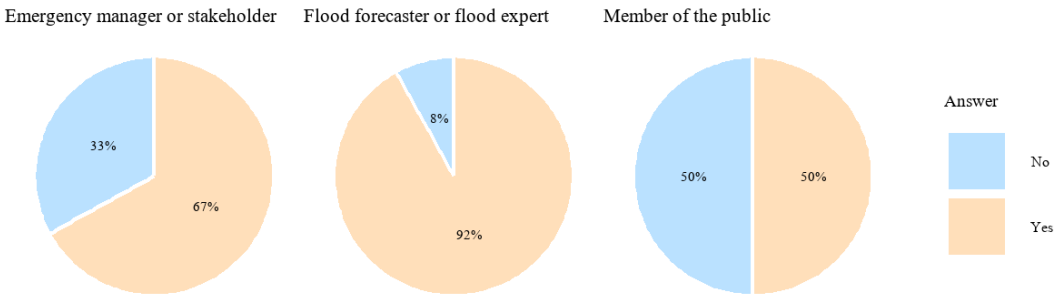


Figure 24: Fifth question of the survey and results obtained - Detail vs. Simplicity in Impact Messages

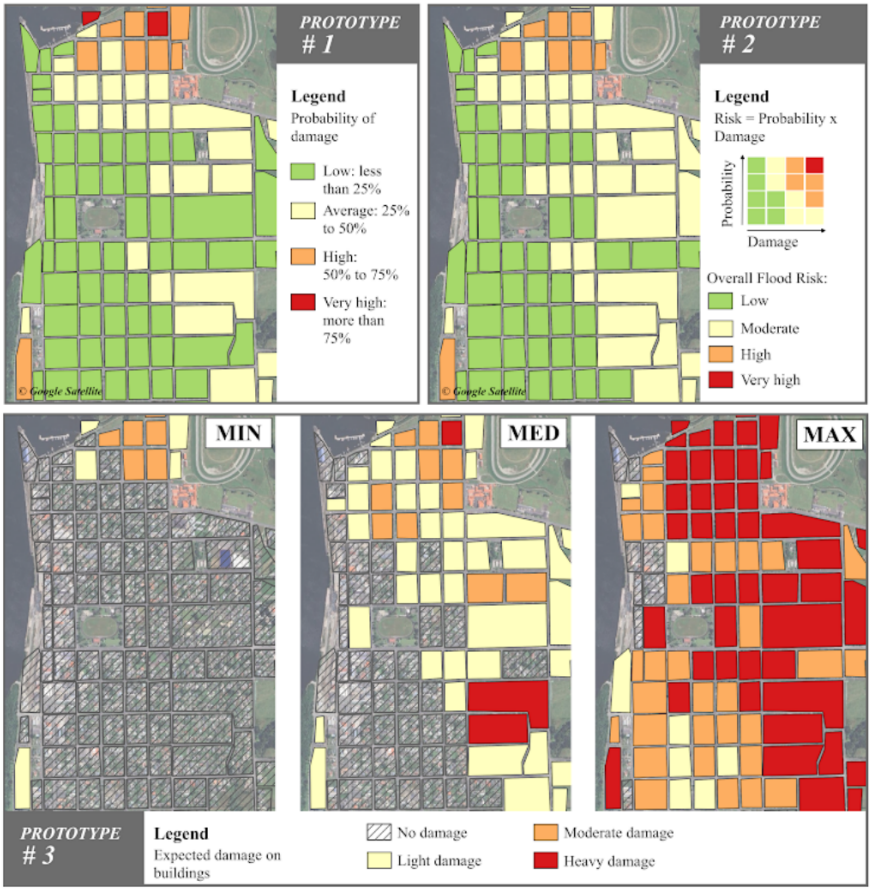
8.1.6 Representation of ensemble-based impact forecasts

Representation of ensemble-based impact forecasts

Ensemble forecasts, which offer multiple equally probable scenarios, are considered more valuable than **deterministic forecasts**,
as they better account for uncertainty. However, this added value often
comes at the cost of increased complexity in communication.

Below, we present different prototypes for communicating **ensemble-based impact forecasts**.

Based on the maps below, which prototype do you prefer for understanding and using impact information during a flood warning? *



- ◯ Prototype 1: Shows the probability of any damage occurring to a building — the information is binary (damaged or not), without detail on severity.
- ◯ Prototype 2: Shows the overall risk, combining both the probability and the severity of expected damage.
- ◯ Prototype 3: Shows the expected damage severity for each block under three different scenarios — minimum, median, and maximum.

(Optional): Please explain why you chose this prototype.

Votre réponse

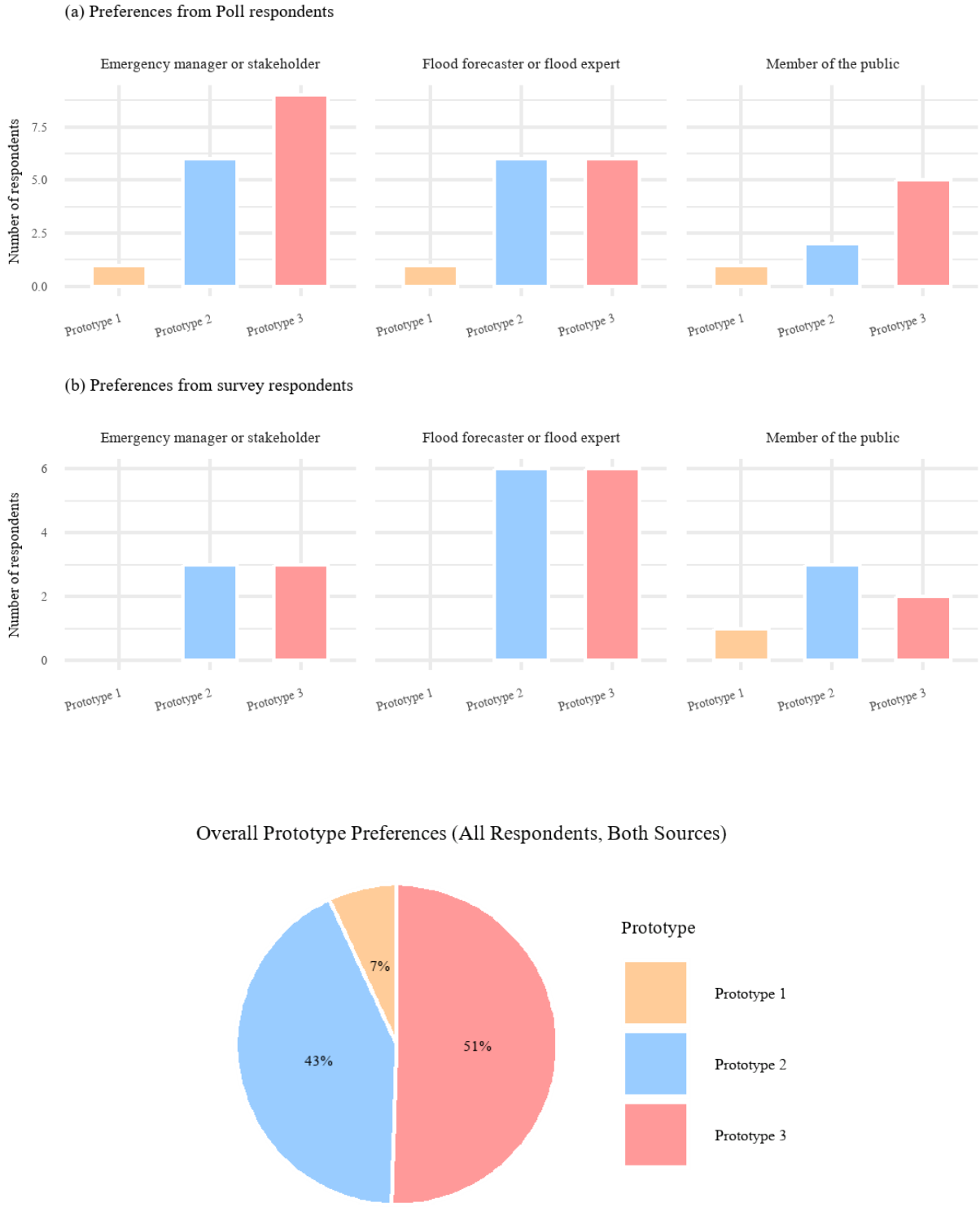


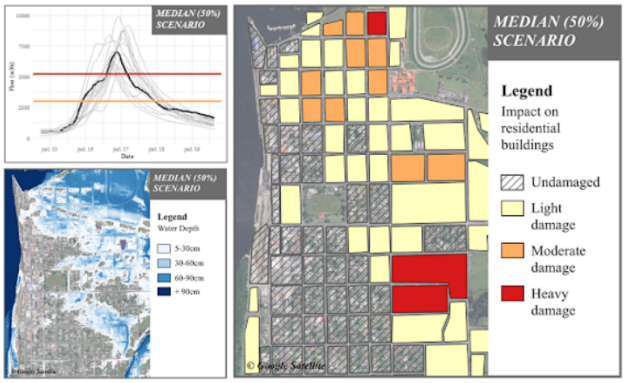
Figure 25: Sixth question of the survey and results obtained - Representation of ensemble-based impact forecasts

8.1.7 Quantity of information

Quantity of information

A flood warning containing only an **impact map** may be insufficient for some users. The **hydrograph** provides essential information about the **timing and duration** of the flood, while the **flood map** offers a view of expected **water depths** in space. However, combining all this information might become overwhelming or redundant, depending on the user's needs.

In your opinion, is the amount of information shown in the figure below appropriate for a flood warning? *



- ◯ Yes, all the elements (hydrograph, flood map, and impact map) are useful and complementary
- ◯ It's too much information — the message becomes confusing or overwhelming
- ◯ Some elements are useful, but others could be removed

(Optional) Do you find any part of the information particularly helpful or unnecessary?

Votre réponse

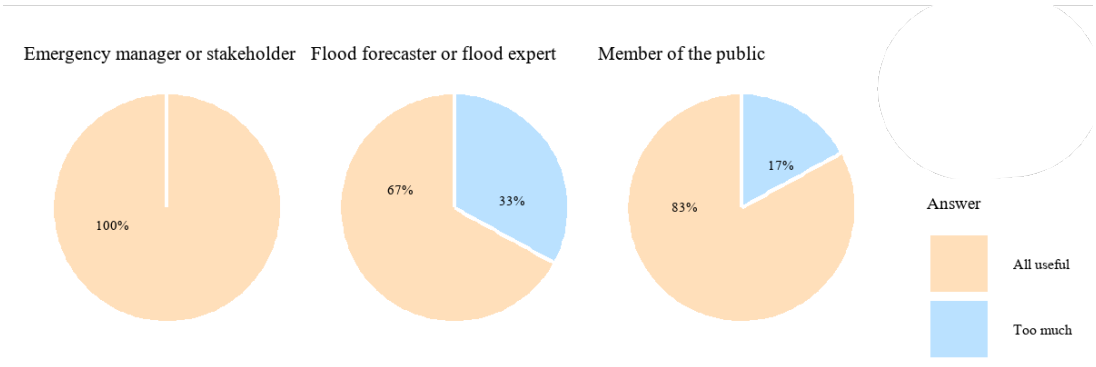


Figure 26: Seventh question of the survey and results obtained - Quantity of information

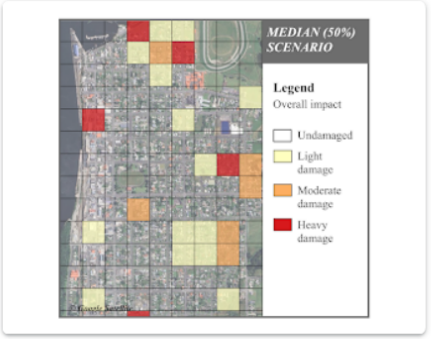
8.1.8 Types of impacts

Types of impacts

In the previous examples, we focused exclusively on **residential buildings**. However, for emergency management and decision-making, knowing the expected impacts on **roads**, **community services**, **utilities**, or **commercial properties** can also be essential — depending on the user's needs.

Which format would you prefer for visualizing this information?

Separate maps for each impact category (e.g., one map for roads, one for services, etc.)



A single map combining all impacts, where categories are integrated and weighted based on their relative importance

(Optional) Can you explain why?

Votre réponse

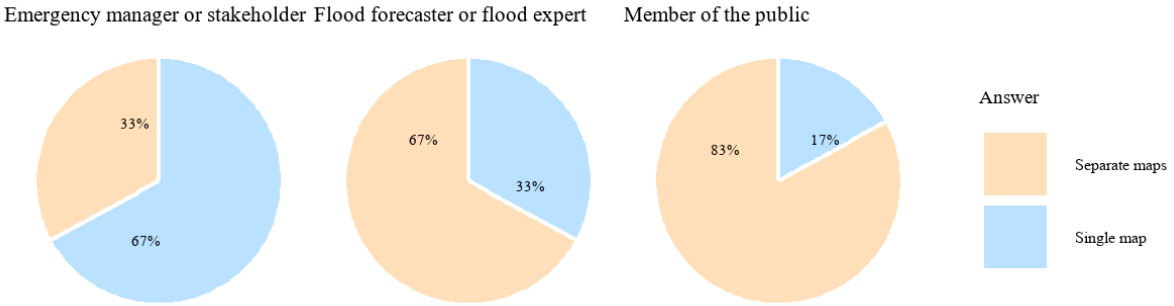


Figure 27: Eighth question of the survey and results obtained - types of impacts

8.2 Analysis of the results of the survey

Profile of respondents

During the webinar, the poll respondents were relatively well distributed across the three categories: members of the public (22%), emergency managers or stakeholders (43%), and flood forecasters or flood experts (35%). Among those who completed the follow-up survey (24 out of the 37 poll respondents), the majority identified as flood forecasters or flood experts (50%), while the remaining respondents were evenly split between the other two categories (25% each).

In the survey, participants were invited to specify their job titles. The "Flood forecaster or flood expert" category included roles such as River/Water Engineer, Catchment Manager, and Civil Engineer. Those identifying as "Emergency manager or stakeholder" included Project Managers, Risk Officers, and Analysts, with affiliations such as Auckland Emergency Management and the Climate Change Commission. The "Member of the public" category was more difficult to interpret, as the webinar targeted a relatively specialized audience. Nevertheless, six respondents selected this category, typically holding positions related to—though not directly involved in—flood response, such as Environmental Planner or Transportation Planner.

The different types of information in a flood warning, and the added value of the impact component

Questions 2, 3, and 7 explored the value of different components of flood forecasts—namely rainfall, flow, inundation, and impacts—to assess how respondents perceived the added value of impact-based information.

In response to the statement *"An impact map provides more understandable, valuable, and actionable information than a water depth map"* (Question 3), a majority of respondents agreed (13 out of 24). Interestingly, three respondents "totally disagreed"—all of whom identified as "Members of the public." This result is somewhat surprising, given the common assumption that impact maps are more accessible to non-experts. Conversely, disagreement was lowest among "Emergency managers or stakeholders" (1 out of 6), which aligns with expectations.

When asked to identify the most important type of information in a flood warning (Question 2), preferences varied across categories (Figure 21). For flood forecasters and experts, rainfall was ranked highest, followed by flow, with inundation and impact information rated similarly. However, for both members of the public and emergency managers/stakeholders, inundation information was considered most important, followed by impact, then flow and rainfall.

Regarding whether presenting all three information types—hydrographs, flood maps, and impact maps—in a warning would be excessive (Question 7), most respondents agreed that the amount of information was appropriate. Nevertheless, 33% of flood forecasters or experts (4 out of 12) disagreed. Of those, two elaborated in optional comments: both believed that while flood maps and hydrographs are useful for experts or emergency managers, the general public would benefit more from a simplified message—preferably just the impact map, complemented by clear text describing timing and duration (rather than a hydrograph). One also suggested using the space to show minimum and maximum impact scenarios.

Interestingly, some respondents identifying as "Members of the public" expressed appreciation for flood maps, as they could easily relate them to their own location and assess personal risk, including danger to life based on water depth.

All emergency managers and stakeholders found the level of information appropriate. One noted that including multiple components helps communicate effectively with diverse audiences. Two respondents highlighted the value of hydrographs over time, and another suggested adding historical context to the hydrograph for non-technical users. Finally, one respondent emphasized that, while building-focused impact maps are useful, they should be complemented with information about infrastructure impacts (e.g., on roads).

Challenges in communicating ensemble-based impact forecasts, managing multiple impact types and damage categories

This topic was explored in the final question of the survey, which asked respondents to choose between two approaches for presenting impact forecasts. The first option consisted of a single, integrated impact map combining various types of impacts (e.g., on buildings, roads, and services), with each type weighted according to its relative importance. This aggregated approach, adopted by Saint-Martin et al. [2016] in a Mediterranean case study, is widely used in the risk assessment community but remains less common in real-time forecasting contexts.

The second option proposed providing separate maps for each impact category. While this approach results in a larger volume of information, it is more reflective of the diversity of on-the-ground impacts.

Respondents' preferences varied by category. Members of the public and flood forecasters largely favoured the "separate maps" approach (83% and 67% respectively), whereas emergency managers and stakeholders tended to prefer the "single map" option. One emergency management respondent justified this choice by emphasizing the need to limit the number of documents consulted during an emergency. Another indicated that the integrated map was easier to interpret.

However, two other emergency managers/stakeholders preferred the "separate maps" approach, arguing that New Zealand's emergency response is multi-agency (e.g., local councils and NZTA for roads, utility companies for electricity and fibre), and that each agency requires information specific to their domain. This view was supported by several flood forecasters who also favoured disaggregated impact maps. Members of the public who preferred this approach commonly cited that it felt more tailored and allowed them to more clearly identify the consequences relevant to their situation.

Even when communicating impact forecasts focused on a single impact category (e.g. buildings), there are many ways of providing information. The easiest, and usually safest way is to provided binary information: a simple "affected / not affected" information for each building or area. Another option, which requires an additional level of precision, consists in providing different damage categories, e.g. "light"/"moderate"/"heavy damage". Question 24 asked the end-users whether they found this extra information useful, even at the cost of greater errors. The greatest agreement to usefulness was found with flood forecasters and flood experts (97%), then emrgency managers and stakeholders (67%), and member of the public were divided (50-50 %).

Finally, beyond the questions of which impact types to communicate and the level of detail to provide, representing impact-based forecasts becomes more complex when dealing with ensemble forecasts. Question 6 specifically addressed this issue by presenting respondents with three prototypes of ensemble-based impact forecasts, each based on a different communication approach:

- Prototype 1 showed the probability of an area being affected—based on a binary outcome. This is the most straightforward method for representing ensemble forecasts in a threshold-based framework: the probability of damage is calculated as the proportion of ensemble members forecasting damage for a given building or area.
- Prototype 2 was based on the Flood Risk Matrix used in the UK to communicate impact-based forecasts [Cole et al., 2016], combining both the probability and the severity of damage.
- Prototype 3 illustrated uncertainty through min / median / max scenarios, without explicitly using probabilities. This approach was previously shown to be the most preferred for communicating flood maps [Jean et al., 2023].

This question was posed in both the live poll during the webinar and the longer survey. When combining responses from both sources across all respondent categories, a slight preference emerged for Prototype 3 (51%), confirming the earlier findings of Jean et al. [2023]. Prototype 2 followed closely behind (43%), while Prototype 1 was the least favored (7%).

Figures 25a and 25b show that preferences were fairly consistent across respondent categories. Flood forecasters and experts were equally split between Prototypes 2 and 3, whereas both emergency managers/stakeholders and members of the public showed a preference for Prototype 3. This again supports previous findings that probability-based representations are often misunderstood by non-technical audiences.

Among those who provided additional comments, respondents favouring Prototype 2 emphasized its clarity for the general public, the completeness of the information, and its usefulness in making critical decisions such as evacuation. Those who preferred Prototype 3 highlighted that it better reflects uncertainty, is easier to understand without a technical background, is familiar to some users (particularly flood forecasters), and offers more complete and visually intuitive information.

Trade-off between Accuracy and Familiarity

Although less directly linked to the previous questions, the trade-off between accuracy and familiarity in impact-based forecasts represents a key challenge in risk communication. Question 4 explored this issue by asking respondents whether they would prefer inundation and/or impact maps based on the closest known scenario (i.e., a map they may already be familiar with and have potentially practiced with), or a map based on the most accurate forecast available, even if it results in unfamiliar outputs.

French perspective: preference for known scenarios

This question resonates with the French experience from the ANR PICS project, where practitioners expressed a clear preference for a scenario-based approach. During group discussions with researchers, many highlighted the importance of using pre-established maps that could serve as references for predefined actions and planning. In contrast, initial feedback from New Zealand practitioners indicated some reluctance to rely on forecasts. However, as collaboration with researchers progressed, trust in the forecasting process grew, and the perceived value of accurate, even unfamiliar, forecasts increased accordingly.

Survey results aligned with this New Zealand perspective: respondents overall preferred to receive maps based on the most accurate forecast, even if they were unfamiliar. This preference was particularly strong among members of the public and emergency managers/stakeholders, suggesting a growing willingness to prioritise precision and up-to-date information over pre-existing familiarity.

9 Conclusions and perspectives

This internship enabled the application of an impact-based flood forecasting routine—similar to that developed during the PhD in France—to a New Zealand case study: the July 2021 flood event in Westport. By adapting an ensemble forecasting chain using NIWA's tools and data (including NZens rainfall forecasts, TopNet hydrological simulations, a precomputed flood library, and the RiskScape software), the study assessed both the potential and limitations of integrating real-time impact information into flood warnings.

Validation against placard data and insurance records helped to evaluate the contribution of downstream forecast components, investigate how uncertainty propagates through the modelling chain, and analyse the correlations between errors at different stages. In parallel, a user survey provided valuable insights into how impact-based products are interpreted and used by different audiences.

This case study offered a rare opportunity for high-resolution validation of impact forecasts and enabled the implementation of a full "cascading uncertainty" analysis across the entire forecasting chain—from rainfall to impacts—something that had not previously been attempted.

This conclusion highlights only the main points addressed in the report and is not intended to be exhaustive.

Several avenues of improvement and future research emerged from this work:

- Validation methods: Introduce additional metrics such as the Gerrity Skill Score (GSS) to assess multi-category impact forecasts.
- Variable thresholds: Explore the integration of building-specific floor height data to replace the fixed 50 cm water depth threshold currently used for damage detection, allowing for more accurate impact prediction.
- Transfer of user survey to France: Replicating the user survey with French stakeholders (including civil protection and the general public) could provide comparative insights into cross-cultural differences in how impact forecasts are perceived and used.
- Additional Westport case studies: Incorporate the February 2022 and August 2022 Westport flood events. Although impacts were limited, data on 6 placarded buildings in February could be used for partial validation. Proxy flood maps for both events could be derived by simulating inundation based on observed flows at Te Kuha.
- Cross-country analysis of cascading uncertainties: Apply the same correlation and divergence analysis used in this study to past French events (e.g., Argens 2010, Alpes-Maritimes 2015, Aude 2018). While PIAF-EPS forecasts are unavailable for these dates, a similar analysis could be performed on the November–December 2019 event, for which observed flood maps and impact data were collected.

These perspectives reflect both elements that were initially planned during the internship but could not be completed due to time or data constraints, and potential next steps that can reasonably be undertaken after the internship and before the end of the PhD. These actions aim to support the preparation of a potential publication based on this work.

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